A U–Pb geochronological review of the Proterozoic history of the eastern Grenville Province

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Abstract: The geological evolution of the eastern Grenville Province can be subdivided into three stages. During the *first stage*, namely pre-Labradorian (> 1710 Ma) and Labradorian (1710–1600 Ma) events, a continental-marginal basin was created and subsequently destroyed during accretion of a magmatic arc formed over a south-dipping subduction zone. Subduction was short-lived and arrested, leading to a passive continental margin. The *second stage* addresses events between 1600 and 1230 Ma. The passive margin lasted until 1520 Ma, following which a continental-margin arc was constructed during Pinwarian (1520–1460 Ma) orogenesis. Elsonian (1460–1230 Ma) distal-inboard, mafic and anorthositic magmatism, decreasing in age northward, is explained by funnelled flat subduction, possibly associated with an overridden spreading centre. As the leading edge of the lower plate advanced, it was forced beneath the Paleoproterozoic Torngat orogen root between the Archean Superior and North Atlantic cratons, achieving its limit of penetration by 1290 Ma. Static north-northeast-trending rifting then ensued, with mafic magmatism flanked by felsic products to the north and south. Far-field orogenic effects heralded the *third stage*, lasting from 1230 to 955 Ma. Until 1180 Ma, the eastern Grenville Province was under the distal, mild influence of Elzevirian orogenesis. From 1180 to 1120 Ma, mafic and anorthositic magmatism occurred, attributed to back-arc tectonism inboard of a post-Elzevirian Laurentian margin. Quiescence then prevailed until Grenvillian (1080–980 Ma) continent–continent collision. Grenvillian orogenesis peaked in different places at different times as thrusting released stress, thereby precipitating its shift elsewhere (pressure-point orogenesis). High-grade metamorphism, thrusting and minor magmatism characterized the Exterior Thrust Zone, in contrast to voluminous magmatism in the Interior Magmatic Belt. Following final deformation, early posttectonic anorthositic–alkalic–mafic magmatism (985–975 Ma) and late posttectonic monzonitic–syenite–granite magmatism (975–955 Ma) brought the active geological evolution of this region to a close.

Résumé : L'évolution géologique de la Province de Grenville orientale peut être subdivisée en trois phases. Au cours de la *première phase*, soit les événements du pré-Labradorien (>1710 Ma) et du Labradorien (1710–1600 Ma), un bassin de marge continentale a été crée et par la suite détruit au cours de l'accrétion d'un arc magmatique formé au-dessus d'une zone de subduction à pendage vers le sud. La subduction n'aura été que de courte durée et elle s'est arrêtée, conduisant à une marge continentale passive. La *seconde phase* concerne les événements entre 1600 et 1230 Ma. La marge passive a duré jusqu'à 1520 Ma, après quoi un arc de marge continentale a été construit durant le cycle orogénique pinwarien (1520–1460 Ma). Le magmatisme elsonien (1460–1230 Ma) distal-intérieur, mafique et anorthositique, dont l'âge décroît vers le nord, est expliqué par une subduction plane concentrique, possiblement associée au recouvrement d'un centre en étalement. Au fur et à mesure que le front de la plaque inférieure avançait, il était forcé en dessous de la racine de l'orogène du Torngat (Paléoprotérozoïque) entre les cratons de l'Archéen supérieur et de l'Atlantique Nord, atteignant sa limite de pénétration vers 1290 Ma. Il y a ensuite eu une dérive vers le nord-nord-est avec du magmatisme mafique flanqué de produits felsiques au nord et au sud. Des effets orogéniques d'arrière champ ont annoncé la *troisième phase*, laquelle a duré de 1230 à 955 Ma. Jusqu'à 1180 Ma, la Province de Grenville orientale subissait l' influence légère et distale de l'orogène elzevirien. De 1180 à 1120 Ma, il s'est produit du magmatisme anorthositique attribué à de la tectonique d'arrière-arc à l'intérieur de la marge de la Laurentia post-elzevirienne. Il y a ensuite eu une période d'accalmie jusqu'à la collision grenvillienne continent à continent (1080–980 Ma). Le cycle orogénique grenvillien a atteint son paroxysme à divers endroits à des temps différents alors que le chevauchement relâchait les contraintes, déplaçant ainsi son accent ailleurs (orogène de point de pression). Du métamorphisme extrême, des chevauchements et un peu de magmatisme caractérisent la zone de chevauchement externe, ce qui contraste beaucoup avec le magmatisme volumineux de la ceinture magmatique interne. Après une déformation finale, du magmatisme précoce post-tectonique anorthositique–alcalin–mafique (985–975 Ma) et du magmatisme tardif

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post-tectonique monzonitique–syénite–granite (975–955 Ma) ont mis fin à l'évolution géologique active de cette région.

[Traduit par la Rédaction]

Introduction

Contained herein and in a companion contribution (Gower and Krogh, in press) is a review of the geological history of the eastern Grenville Province (defined as in Fig. 1) as seen through U–Pb geochronological data, tempered with detailed familiarity with current concepts for the geological make-up of the region. Our foremost objective is to evaluate all published U–Pb geochronological data for the region and to provide a critiqued geological interpretation of what they mean. We are acutely concerned that data are commonly misused, used in an uncritical manner, or taken out of context and that these mistakes have become entrenched in the literature through iteration, especially due to quoting secondary sources. Also, with the increasing ease of manipulating databases, there is likely to be increasing neglect of individual treatment of information. All of the data reported has meaning in one sense or another, but only by detailed examination of results on an individual basis, then relating them to the whole database is it possible to determine whether the best interpretation has been offered.

In carrying out this review, we have evaluated and quoted original, "final" sources, although we have used preliminary information if nothing more is available. We have tried to be especially circumspect in the latter situation as we regard the reporting of dates in "preliminary" form to be an unfortunate practice that can cause long-term confusion. In addition, we feel that in some cases, to understand the geochronological data it is necessary to "read between the lines." We have not hesitated to express any doubts when there seems some reason to challenge results as originally presented. Where we have offered interpretations of geochronological data different from those of the original authors, we have tried to make these instances clear.

This review of the Proterozoic evolution of the eastern Grenville Province is presented in historical format, divided into three parts, subtitled "arrested subduction," "funnelled flat subduction," and "pressure-point orogenesis" to highlight a key tectonic feature of each stage. These correspond in time to pre-1600 Ma (but mostly post-1800 Ma), 1600–1230 Ma, and 1230–955 Ma (Fig. 2). At the end of each stage, a tectonic interpretation is offered. An attempt has been made to keep anachronism in its place, but it must be remembered that this region has been subjected to repeated geological activity, thus it is inescapable that some events must be either harkened back to or anticipated. Our "unit" of geographical reference in the eastern Grenville Province is a terrane (Fig. 1), individual place names having been largely omitted. What terranes mean in tectonic terms when applied to the eastern Grenville Province has been addressed elsewhere (e.g., Gower 1996). Note that we have adopted names for units that are currently most commonly in use, as it is not our purpose to standardize or refine stratigraphic nomenclature.

Part I. Pre-Labradorian and Labradorian (arrested subduction)

The pre-Labradorian and Labradorian U–Pb geochronological database for the eastern Grenville Province and its interpretation have been evaluated in detail by Gower and Krogh (in press). To avoid a serious gap in the geological history of eastern Laurentia that would otherwise be present in this volume, a brief review is presented here. Excluded from this summary are detailed reporting of ages obtained, their precision and location, bibliographical sources, discussion of the shortcomings of specific results, or historical review of the various models that have been proposed, all of which has been given by Gower and Krogh (in press). The distribution and main lithological subdivisions of these rocks are given in Fig. 3.

Pre-Labradorian

Pre-Labradorian rocks can be sub-divided into three groups, namely (*i*) those representing the southern extension of the foreland into the parautochthonous northern fringe of the Grenville Province, (ii) mainly granitoid orthogneisses in the interior Grenville Province, and (*iii*) high-grade metasedimentary gneisses. The parautochthonous group includes the early Paleoproterozoic Kaniapiskau Supergroup in western Labrador and adjacent Quebec, and Archean and Paleoproterozoic rocks extending southward into the Grenvillian parautochthon from the eastern Churchill and Makkovik provinces. The granitoid orthogneiss (with minor metasedimentary gneiss) occur in the Mealy Mountains terrane, and have given ages between 1782 and 1712 Ma. The high-grade metasedimentary gneisses are mostly pelitic to semipelitic rocks with minor quartzite, calc-silicate rocks and amphibolite (derived in part from pillowed mafic volcanic rocks). Most interpretations for this unusually large and homogeneous sedimentary package envisage some form of subaqueous environment distant from high-energy sources of detritus supply, but oxidized shale and a lateritic origin have also been considered. The age of the rocks is only constrained between 1805 and 1709 Ma, having an Archean and Paleoproterozoic provenance from farther north in eastern Laurentia.

Labradorian

Early Labradorian

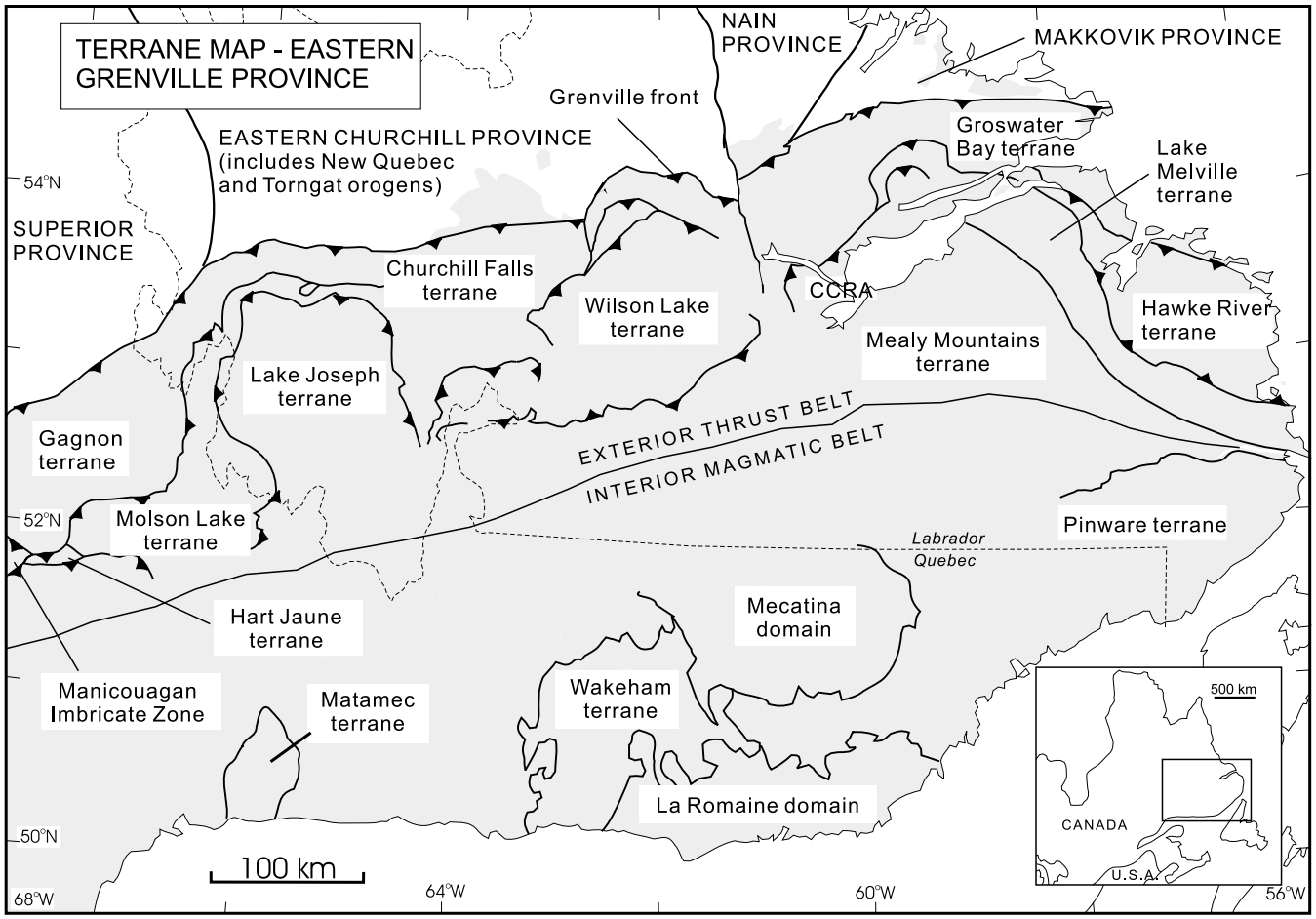
Early Labradorian events are taken here to include those between ca. 1710 and 1680 Ma, currently represented only by a 1709 Ma nebulitic granodiorite from eastern Labrador.

Calc-alkaline magmatism

Calc-alkaline magmatism (quartz diorite, granodiorite, abundant K-feldspar megacrystic granodiorite to quartz monzonite, minor tonalite and granite) related to Labradorian orogenesis occurred between 1680 and 1650 Ma and has

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Fig. 1. Definition of the eastern Grenville Province, also showing distribution of terranes and subdivision into Exterior Thrust Belt and Interior Magmatic Belt. CCRA, Cape Caribou River allochthon.



been subdivided into three age groups, which formed at 1677 Ma (Neveisik Island event), 1671 Ma (Earl Island event), and 1658–1649 Ma (Double Island event) (cf. fig. 6 of Gower et al. 1992). These groups correlate with the Lake Melville, Hawke River, and Groswater Bay terranes, respectively, and suggest a northward decrease in age of plutonism. The rocks are strongly foliated or gneissic, consistent with emplacement before, or during, major Labradorian deformation and magmatism.

Pre-Trans-Labradorian metamorphism and deformation

A considerable body of remarkably consistent evidence for deformation and metamorphism between 1665 and 1655 Ma requires that a radical change in tectonic conditions occurred. The timing of metamorphism and deformation have been determined by bracketing ages, dating melts, dating inferred syntectonic intrusions, and dating minerals interpreted to have been directly formed during metamorphism. The single dated exception involves a pegmatite in the Rigolet area, considered by Corrigan et al. (2000) to have been emplaced syntectonically with mylonitization at 1619 Ma. Gower and Krogh (in press) offer an alternative interpretation whereby mylonitization truncated by the pegmatite occurred between 1665 and 1655 Ma, and the deformation affecting the pegmatite was unrelated and (much?) younger.

Trans-Labrador magmatism

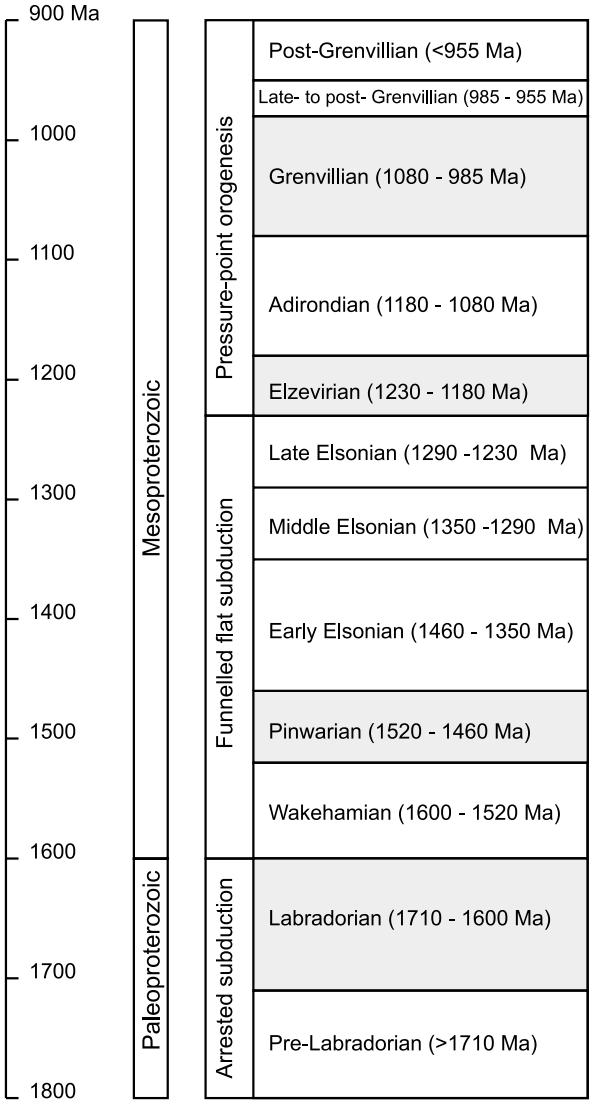
Three groups of rocks are addressed under this heading. These are (*i*) Trans-Labrador batholith, (*ii*) coeval supracrustal units, and (*iii*) coeval granitoid rocks in the interior part of the eastern Grenville Province.

The Trans-Labrador batholith forms a linear belt at least 600 km long and less than 100 km wide at the northern margin of the former Labrador orogen and was emplaced between 1654 and 1646 Ma. The Trans-Labrador batholith is slightly oblique to the Grenville front, the northern edge being 50 km north of the Grenville front in eastern Labrador, but about 50 km south of the Grenville front in western Labrador. Rock types are dominated by equigranular to megacrystic quartz monzonite, monzogranite, monzosyenite, syenite, and granite. The chemical character is transitional calc-alkaline to alkali-calcic.

Integral with Trans-Labrador batholith magmatism are supracrustal sequences (felsic volcanic rocks and related sediments) situated along its northern fringe. These have been interpreted as the extrusive carapace to the Trans-Labrador batholith, but preserved only on the northern side because of progressively deeper exhumation from north to south linked to Grenvillian thrusting. The supracrustal sequences are termed, from west to east, the Blueberry Lake, Mackenzie Lake, and Bruce River groups. Other possibly correlative units, although

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Fig. 2. Time subdivisions used for the eastern Grenville Province. The main periods of orogenesis are shaded. Elzevirian orogenesis is less convincingly orogenic than other periods.



not all dated, include felsic volcanic rocks south of Mount Benedict and in the Groswater Bay terrane of eastern Labrador, the Sims Formation in western Labrador, and supracrustal rocks in the Pinware terrane (Pitts Harbour Group). The Pitts Harbour Group consists mostly of fine-grained, recrystallized, banded quartzofeldspathic rocks, locally showing inhomogeneous texture. The rocks have been interpreted as felsic volcanoclastic, but this is equivocal, and even the assertion that they are supracrustal relies on common association with rocks of more certain supracrustal protolith.

Evidence exists for coeval granitoid rocks south of the Trans-Labrador batholith in western and central Labrador, but is somewhat equivocal (discussed by Gower and Krogh, in press). More rigorous are Trans-Labradorian ages from the Pinware terrane from two gneissic quartz monzonites. Inheritance of similar age in younger rocks has also been reported elsewhere in southernmost Labrador.

Trimodal mafic–anorthositic–monzogranitic magmatism

Rocks included under this heading were emplaced between 1650 and 1620 Ma, overlapping with both the preceding Trans-Labradorian magmatism and events considered in the next section. It cannot be stressed too much that the history of the Trans-Labrador batholith and the trimodal mafic–anorthositic–monzogranitic magmatism are inextricably linked temporally, spatially, and compositionally. Rocks can be divided into three groups; mafic, anorthositic, and monzogranitic, which also have spatial and temporal significance.

Rocks included in the mafic group are hydrous, commonly layered intrusions. With the exception of the Adlavik intrusion, the rocks are most abundant where associated with the highest grade host rocks, suggesting an association between deeper crustal levels and high-volume mafic magmatism. The Adlavik intrusion contrasts from other bodies mentioned below as it is in the Makkovik Province, is on the north side of the Trans-Labrador batholith, and is the oldest. It may not be unique, however, as a coeval and lithologically similar body occupies a comparable tectonic setting in the eastern Churchill Province east of Michikamau Lake. South of the Trans-Labrador batholith, dated rocks include the Grady Island layered mafic intrusion and White Bear Arm complex in eastern Labrador and the Ossok Mountain Intrusive Suite in western Labrador, but there are many other, as yet undated, almost certainly correlative intrusions throughout the northern half of the eastern Grenville Province.

Anorthositic magmatism is represented by the Mealy Mountains Intrusive Suite, Northwest River anorthosite, and Alexis River anorthosite, as well as smaller unnamed bodies. These rocks are, broadly speaking, situated interior of the layered mafic intrusions. The Northwest River anorthosite forms the upper part of the Cape Caribou River allochthon and can be regarded as a tectonically isolated part of the Mealy Mountains Intrusive Suite.

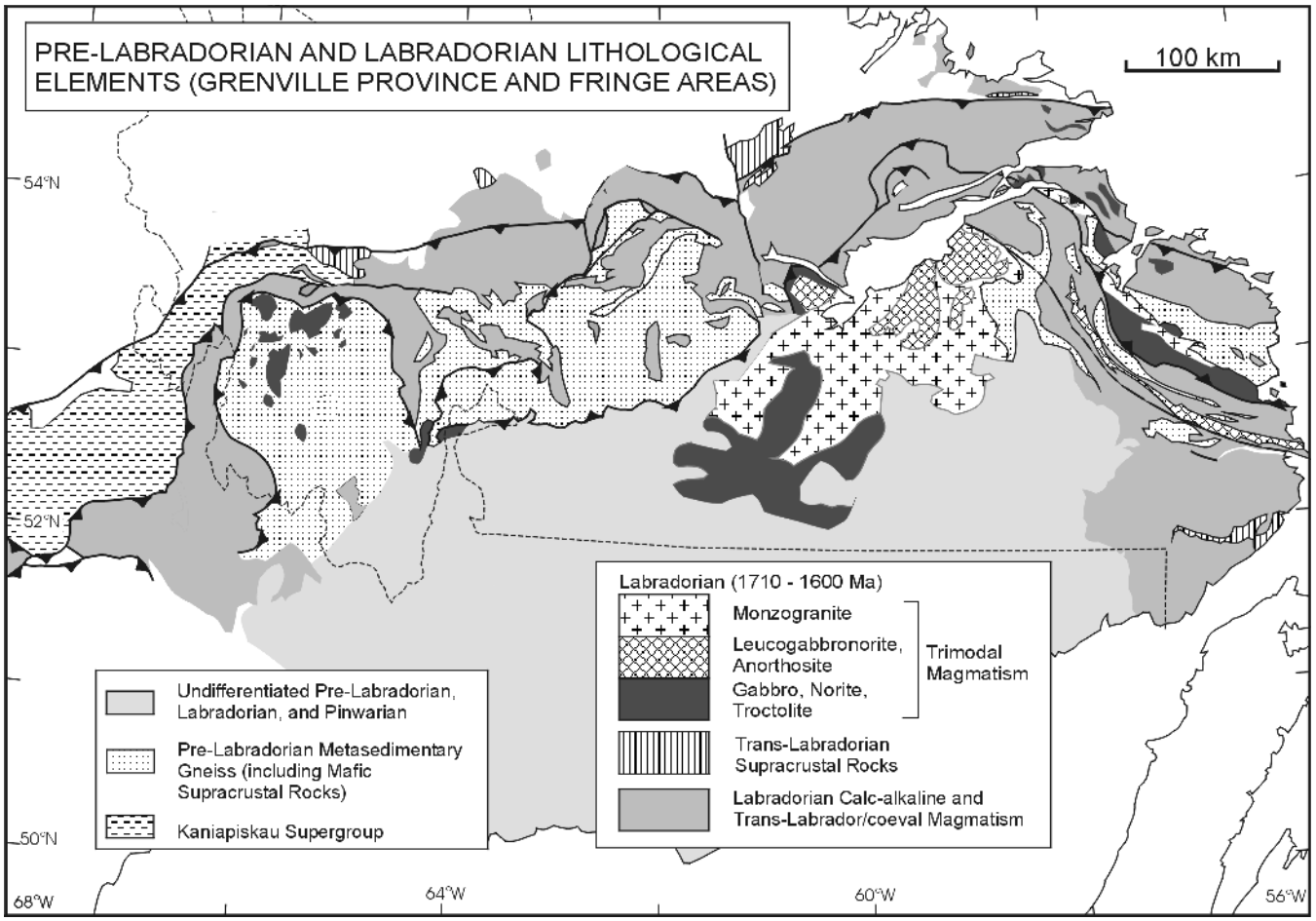
Monzogranitic magmatism covers the compositional range from monzonite to granite, although monzonite is volumetrically dominant. Some rocks have been termed syenite, but such a designation is not intended to imply an “alkalic” tectonic context. Association between monzogranitic and mafic rocks ranges from intimate-complex to mutually independent. The monzogranitic rocks are normally separate from anorthositic units, although it is not uncommon to find enclaves of anorthosite in monzonite (but rarely the reverse). In some cases, the age of intimately associated monzonite has been taken as dating associated mafic rocks (e.g., Adlavik Intrusive Suite and Grady Island intrusion). Some granitic bodies emplaced at about the same time do not have apparent spatial links to mafic or anorthositic rocks; examples are given by Gower and Krogh (in press).

Post-Trans-Labradorian deformation and metamorphism

Post-Trans-Labradorian deformation and metamorphism are distinguished from earlier deformation in that they extended over a longer period, were more sporadic, and were less intense. They involved migmatization, metamorphism and amphibolite-facies retrogression, and occurred in-parallel with the earlier reviewed trimodal magmatism.

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Fig. 3. Major pre-Labradorian and Labradorian lithological elements.



Late Labradorian

Late magmatism, between 1625 and 1600 Ma, involved the emplacement of small plutons, stocks and minor granitic intrusions, most of which were intruded at the beginning of this final phase of Labradorian activity, probably representing nothing more than the dwindling manifestation of trimodal mafic–anorthositic–monzogranitic magmatism. Evidence for late Labradorian deformation and metamorphism relies on a rather disparate set of data, reviewed by Gower and Krogh (in press). Taken at face value, they indicate melting in the Lac Joseph terrane and Cape Caribou River allochthon, mylonitization at the Groswater Bay – Lake Melville terrane boundary, mafic dyke injection in the Hawke River terrane, and lower temperature closure in the Churchill Falls terrane. Events were sporadic, diverse, and mostly minor.

Tectonic model

Pre-Labradorian (>1710 Ma)

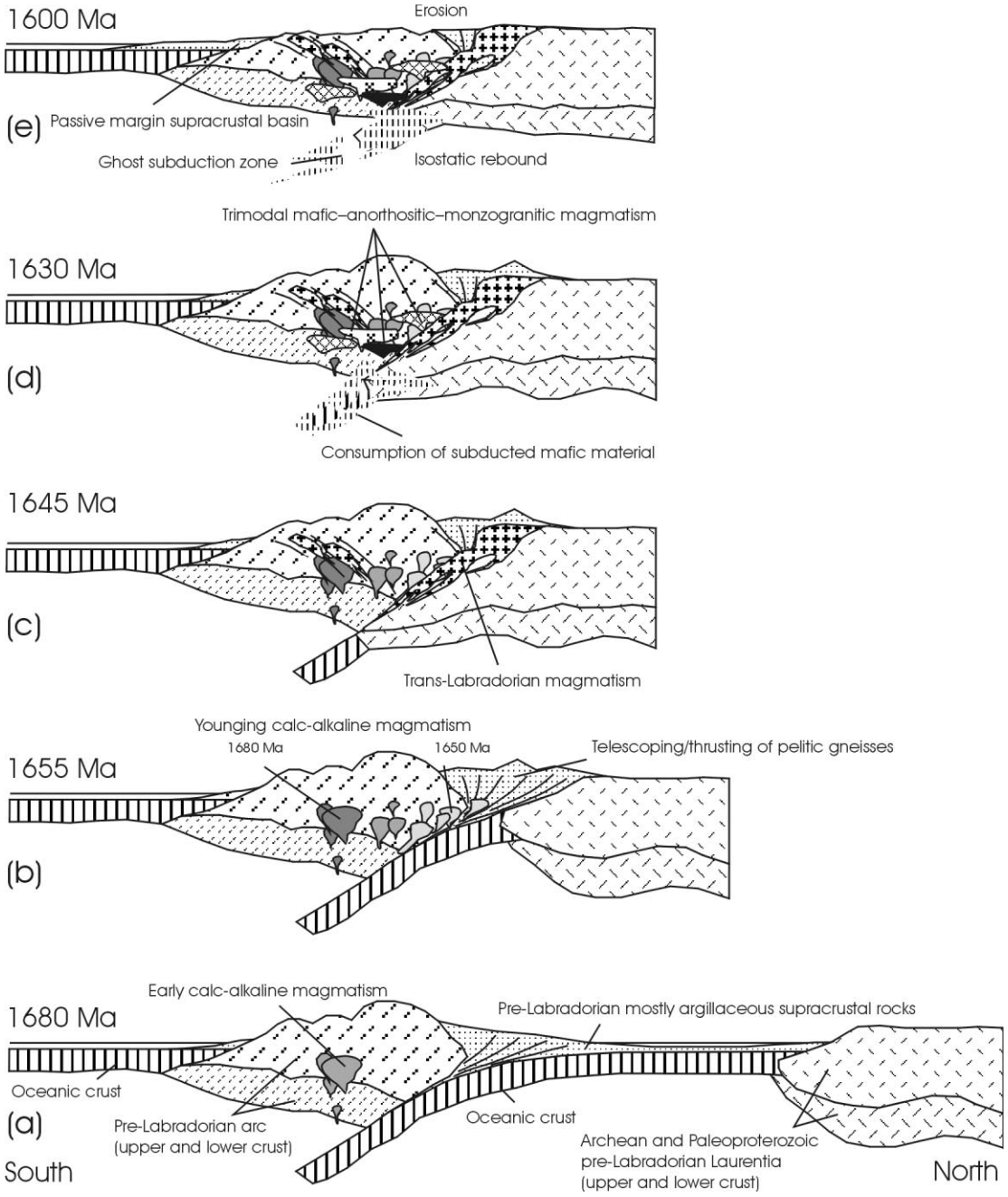
Sparsity of data prevents meaningful assessment of a potential tectonic environment for pre-Labradorian granitoid rocks at this stage. With respect to the widely distributed metasedimentary gneiss, although acknowledging a pre-Labradorian Laurentian contribution, Sr, Nd, and Pb studies have collectively failed to demonstrate any conclusive evidence for involvement of an appreciable volume of old crustal material, thus reinforcing conclusions that deposition of the supracrustal rocks occurred after ca. 1800 Ma from local juvenile Paleoproterozoic sources. Note that correlation can be made with the post-1792 Ma time of sedimentation in the psammite and pelite zones in the Ketilidian Mobile Belt (Garde et al. 1998), thus providing an elegant broader regional context in which to evaluate pre-Labradorian history in the interior eastern Grenville Province (cf. fig. 4*a* of Gower 1996).

Labradorian (between 1710 and 1655 Ma)

It has been argued (cf. Gower and Krogh, in press, for references) that the calc-alkaline rocks formed in an arc that was outboard of pre-Labradorian Laurentia and over southward-subducting oceanic crust (Fig. 4). Southward subduction explains (*i*) lack of pre-1655 Ma rocks in pre-Labradorian Laurentia farther north, (*ii*) lack of contamination of Labradorian arc-related rocks by an appreciably older source, (*iii*) progressive younging of calc-alkaline rocks from south to north, being steadily accreted to the front of the arc, and (*iv*) south-dipping mantle reflectors in the southern Grenville Province, representing the remnants of the subduction zone. One should be heedful that evidence of pre-1655 Ma rocks north of the Trans-Labrador batholith would impair the above hypothesis, which prompts mention of the Bridges layered mafic intrusion within the Nain Plutonic Suite and dated by Sm–Nd isotopic methods to be

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Fig. 4. Tectonic model for the evolution of the eastern Grenville Province during pre-Labradorian and Labradorian orogenesis.



1667 ± 75 Ma, interpreted as the time of crystallization. Lack of precision prevents exacting use of this date, and, by the same token, given its geological context, clearly identifies it as an important candidate for U–Pb studies.

Collisional orogenesis (1665–1655 Ma)

Metamorphism and deformation between 1665 and 1655 Ma is taken to date the time of arc accretion to Laurentia, the collision initiating Trans-Labrador batholith magmatism (Fig. 4*b*). It seems indisputable that the very high-grade rocks now exposed at the surface were formed during the 1665–1655 Ma deformational and metamorphic event. It must be kept in mind that, although major high-grade mylonite zones attest to major displacement, uplift of these high-grade rocks from lowermost crustal levels need not have been accomplished entirely at this time. Nevertheless, high-grade enclaves within the relatively shallow-level Trans-Labrador batholith (as demonstrated by preservation of its volcanic carapace along its northern fringe), suggest that much of the uplift predated Trans-Labrador magmatism.

Trans-Labradorian magmatism (1654–1646 Ma)

Common ground seems to have been reached that Trans-Labradorian magmatism is the consequence of melting triggered by crustal thickening, although less of a consensus has been achieved on what caused crustal thickening in the first place. Gower and Krogh (in press) assess various proposed models, coming to the conclusion that crustal thickening

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followed on from accretion of Labradorian calc-alkaline arc rocks during the 1665–1655 Ma period of magmatism and deformation, after which time southward-directed subduction ceased (Fig. 4*c*).

Trimodal mafic–anorthositic–monzogranitic magmatism (1650–1620 Ma)

The preferred thesis of Gower and Krogh (in press) is that the mafic–anorthositic–monzogranitic rocks are the later products of the crustal thickening that initially produced Trans-Labradorian magmatism. In this regard, a model proposed by Duchesne et al. (1999) seems applicable, whereby crustal tongues of lower mafic crust reach sufficiently high pressures and temperatures to produce jotunitic melts at 1200°C and high-alumina basalts at 1300°C. The magmas and crystal mushes are channelled by crustal structures, and granitoid rocks are generated by differentiation of the mafic magma and (or) partial melting of the lower crust. In the case of mid- to late-Labradorian magmatism, the tongue of mafic crust could be regarded to be the last of the oceanic crust to go down the subduction zone, plus whatever mafic continental lower crust that tried to follow (Fig. 4*d*). This scenario is consistent with a concentration of post-Trans-Labradorian deformation and metamorphism in more southerly regions.

Late Labradorian (1620–1600 Ma)

The final stage of Labradorian history is a gradual dwindling of activity as the crust regained stability. This led to quiescence and the development of a passive, sediment-accumulating continental margin (Fig. 4*e*).

Part II. Post-Labradorian – Pre-Pinwarian, Pinwarian, and Elsonian (funnelled flat subduction)

Post-Labradorian – Pre-Pinwarian events

Geochronological "evidence"

As pointed out by Gower et al. (1990*b*), and iterated by Gower (1996), one of the major enigmas of Proterozoic evolution in Laurentia is the lack of reliably dated geological events for the period between 1600 and 1520 Ma and the relative sparsity of activity in the 10–20 million years before and after this core period. Reference to Gower and Krogh (in press) will affirm that most late-Labradorian dates are closer to 1620 Ma than to 1600 Ma, for example. The key word is "reliably," as several dates within this period from the eastern Grenville Province have been reported, and, at issue, is judging their significance. All samples from the eastern Grenville Province having nominal U–Pb ages between 1600 and 1520 Ma are suspect, either being very imprecise, and (or) involving long extrapolations from data plotting close to lower intercepts, or projecting to negative lower intercepts.

Wakeham Supergroup

The Wakeham Supergroup is a package of greenschist- to lower amphibolite-facies sedimentary and volcanic rocks in the Lac Wakeham area, 55 km north of Baie-Johan-Beetz (Fig. 5). The group was subsequently elevated to supergroup status by Martignole and Indares (1988) who introduced the names Aguanus and Davy groups for the east and west parts of the sequence, respectively, and which are separated by the Nabisipi mylonite zone. The Aguanus Group was envisaged as the lowest stratigraphically and described as shallow-water arenites intercalated with thick subaerial rhyolitic and minor basaltic rocks of continental tholeiitic to subalkaline affinity in its upper parts. The Davy Group comprises basal conglomerate fans, overlain by siliceous and feldspathic arenites and intruded by numerous mafic sills. Two periods of folding were inferred to have affected the Aguanus Group, whereas only one was seen in the Davy Group (Martignole et al. 1994). Abandonment of the two divisions and return to the term Wakeham Group has been recommended by Madore et al. (1997*a*, 1997*b*), who argued that the divisions lack separate identity and that stratigraphic superposition has not been proved. This suggestion is not followed here for reasons given in the following text.

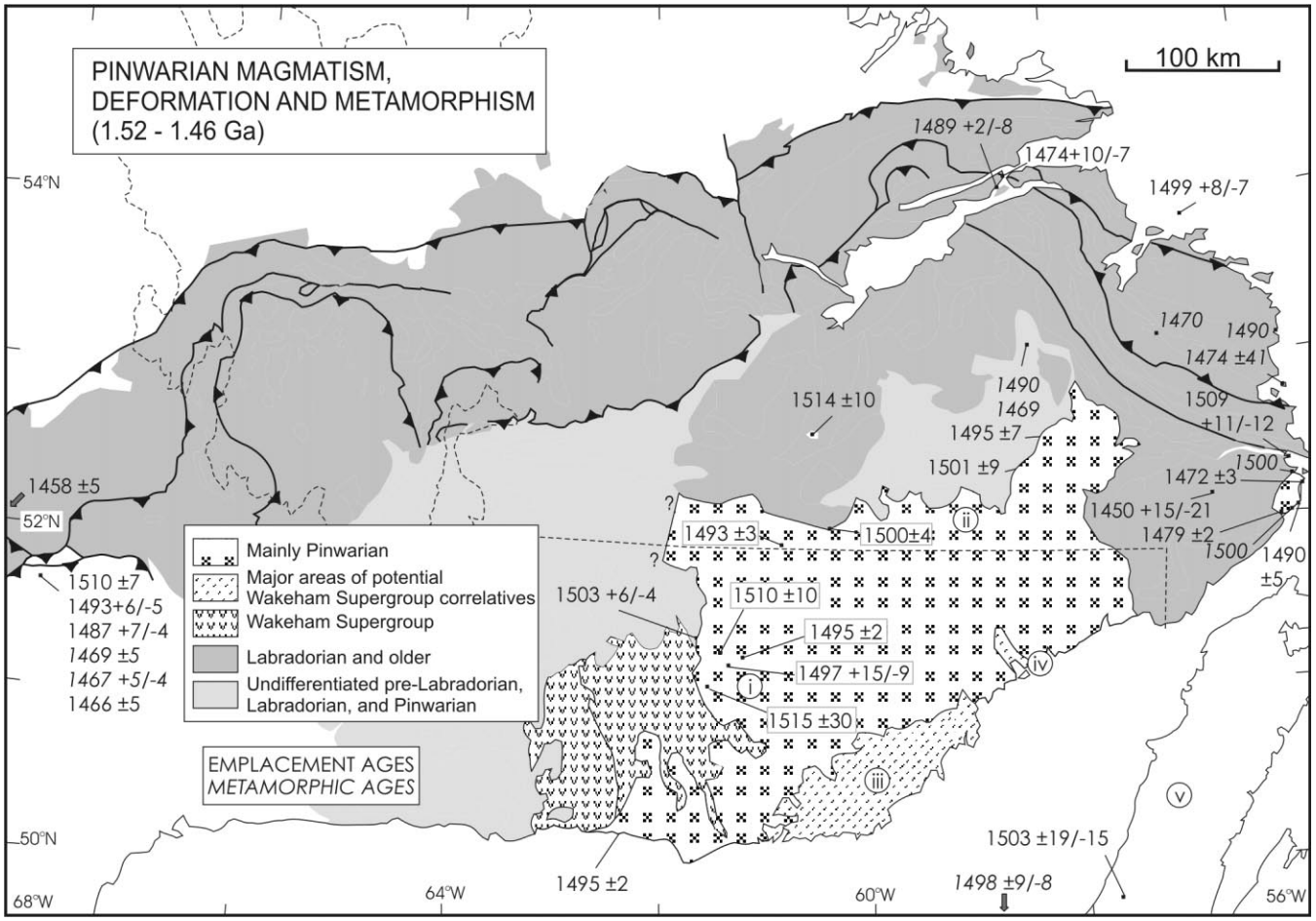
Until recently, the age of the Aguanus Group was based on a 1271^{+13}_{-3} Ma zircon age (Loveridge 1986) from an associated rhyolite, interpreted to be extrusive. The age of the Davy Group was constrained by a 1246 ± 5 Ma zircon date (Clark and Machado 1995) from a feldspar porphyry in the basal part of the sequence. Madore et al. (1997*a*, 1997*b*), however, determined that supracrustal rocks assigned to the eastern part of the Wakeham Supergroup are intruded by a Kataht porphyry dated to be 1503^{+6}_{-4} Ma. Furthermore, U–Pb investigation of detrital zircons from the Wakeham Supergroup failed to find any zircons having ages less than 1550 Ma (in two age peaks — 1600 Ma and 2700 Ma; Larbi et al. 1998, 1999). Sm–Nd data reported by Larbi et al. (1998, 1999) showed ε_{Nd} of 0 to +4 and depleted mantle model ages of 1618–1450 Ma from the northeast Wakeham Supergroup; ε_{Nd} reaching as low as –5 and depleted mantle model ages of 1803–2039 Ma from the southeast Wakeham Supergroup; and ε_{Nd} greater than +4 from the western Wakeham Supergroup. They concluded that the northeastern Wakeham Supergroup had an important contribution from an old, possibly Archean source, whereas, farther south, Proterozoic input became dominant, and that the western Wakeham Supergroup involved an even greater Proterozoic contribution. These data constrain the depositional age of the Wakeham Supergroup to between 1600 and 1510 Ma and require that the previously dated geon 12 porphyries be intrusive rather than extrusive, as had been originally interpreted. Given the contrast in ε_{Nd} between the eastern and western parts of the Wakeham Supergroup, the possibility still remains that two supracrustal sequences of differing age are represented, which would allow reconciliation with the deformation history reported by Martignole et al. (1994). On this basis, it is considered here premature to abandon the terms Aguanus River and Davy groups (as perhaps it was to introduce them originally?).

Potential Wakeham Supergroup correlatives

One of the dilemmas that routinely confronts the field geologist working in rocks that show a range of metamorphic grades is to determine whether the variation is a function of time or space. Although the problem would seem easily resolved by tracing rock units from one area to another, complications commonly arise due to lack of structural, outcrop, or mapping continuity. The issue achieves relevance when addressing potential correlatives of the Wakeham Supergroup outside its type region, as most of these are higher grade

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Fig. 5. Distribution of major areas of post-Labradorian – pre-Pinwarian supracrustal rocks and Pinwarian U–Pb ages. The area labelled as "Mainly Pinwarian" is speculative. (*i*) to (*v*) indentify the areas introduced by the same numbers in the text section Potential Walcehem Supergroup correlatives.



rocks and may be older: some candidates, by no means an exhaustive list, follow.

(*i*) The Boulain complex, immediately east of the Wakeham Supergroup, and for which a 1515 ± 30 Ma metamorphic age has been reported, is described as comprising leucogranitic gneiss, foliated granite, foliated pegmatite and foliated gabbro (Madore et al. 1997*a*, 1997*b*; Verpaelst et al. 1997*a*, 1997*b*). This assemblage would seem to have little in common with the Wakeham Supergroup, but the presence of muscovite (up to 20%) and tourmaline, coupled with morphologically varied zircons having ages between 1.5 and 2.0 Ga (Madore et al., and Verpaelst et al., op. cit.), suggest a partly sedimentary protolith having a similar age to the Wakeham Supergroup.

(*ii*) East of St. Augustin River, Gower (2001) mapped a package of quartz-rich, quartzofeldspathic (psammitic) gneisses associated with quartzite and rare pelitic layers, which differ lithologically from the vast areas of pre-Labradorian pelitic gneiss farther north. The rocks might be correlative with those in the Wakeham Supergroup — a suggestion yet to be tested isotopically.

(*iii*) On the north shore of the St. Lawrence River (Fig. 5), a roughly 600 km^2 area is described by Bourne et al. (1978) as underlain by meta-arkose and feldspathic meta-quartzite or "charnockitic" gneisses (no orthopyroxene was positively identified in the field). The latter rocks, because they are interlayered with quartzite, are also suggested to be of sedimentary origin.

(*iv*) In the Baie de Jacques Cartier area, cordierite-bearing pelitic and semi-pelitic gneisses, locally intercalated with calc-silicate rocks and quartzite, have been investigated by Perreault and Martignole (1988).

(*v*) Screens of metasedimentary gneiss are found sporadically in the Long Range Inlier in western Newfoundland (Owen 1991) and include some unusual gedrite-bearing gneisses that Owen and Greenough (1995) interpret as having been derived from a weathered greywacke. Lack of confirmation of pre-Pinwarian orthogneiss from the region (Heaman et al. 1996), encourages the notion that these supracrustal rocks could be similar in age to the Wakeham Supergroup.

Pinwarian Orogeny

The Pinware terrane was first identified as a separate tectonic entity by Gower et al. (1988), and rocks of Pinwarian age in the eastern Grenville Province were first reported by Tucker and Gower (1990). The term Pinwarian Orogeny was introduced by Tucker and Gower (1994) for rocks having ages between 1500 and 1470 Ma, but subsequent dating has extended these limits, conservatively, to between ca. 1520 and 1460 Ma. In

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this article, events after 1460 Ma are termed Elsonian (see later in the text).

Granitic magmatism

The key feature that characterises dated Pinwarian magmatism is its overwhelmingly granitic (sensu stricto) character, as regards both sizable bodies and minor intrusions (Fig. 5). The larger bodies include (*i*) a 1514 ± 10 Ma granite in the Kenamu River area (James et al. 2000), (*ii*) three Kataht Suite porphyries having ages of 1510 ± 10 Ma, 1503^{+6}_{-4} Ma, and 1495 ± 2 Ma, and a rapakivi granite associated with the Kataht Suite dated to be 1497^{+15}_{-9} Ma (Madore et al. 1997*a*, 1997*b*; Verpaelst et al.1997*a*, 1997*b*), (*iii*) the Upper Paradise River pluton alkali-feldspar granites having ages of 1501 ± 9 Ma and 1495 ± 7 Ma (Wasteneys et al. 1997), (*iv*) an augen gneiss in the southern Wakeham area dated to be 1495 ± 2 Ma (Clark and Machado 1995, who interpreted the date as metamorphic), (*v*) a 1493 ± 3 Ma K-feldspar megacrystic granite and a 1500 ± 4 Ma quartz monzonite in the Fourmont Lake area and most likely part of the Petit Mecatina suite (James et al. 2001), (*vi*) on the coast, the Cape Charles quartz monzonite, St. Peter Bay granite, and Wolf Cove quartz monzonite, having ages of 1490 ± 5 Ma, 1479 ± 2 Ma, and 1472 ± 3 Ma, respectively (Tucker and Gower 1994), (*vii*) granitic gneisses in western Newfoundland having ages of 1503^{+19}_{-15} Ma and 1498^{+8}_{-7} Ma (Heaman et al. 1996; Currie et al. 1992, respectively). An inherited age of 1530 ± 30 Ma from the 964 Ma Riviere Bujeault headwaters quartz syenite (Gower et al. 1991) may be a further indication of Pinwarian material in that area.

Minor dated granitic intrusions are the 1509^{+11}_{-12} Ma Long Harbour granitic vein (Scott et al. 1993), the 1499^{+8}_{-7} Ma Double Island pegmatite (Schärer et al. 1986), the 20-m-wide 1474^{+10}_{-7} Ma Wolfrey granite (Corrigan et al. 2000), and, in the Hart Jaune area, a 1493^{+6}_{-5} Ma granite vein (Scott and Hynes 1994*a*), a 1469 ± 5 Ma granitic sheet, and a 1466 ± 5 Ma granite (Scott and Hynes 1994*b*). A 1510 ± 7 Ma mafic pegmatite from the Hart Jaune area (Gobeil et al. 1996) represents the only compositional departure from granite. A worrisome feature of the seemingly valid data from the Wolfrey granite is that the lower intercept projects to 353^{+130}_{-126} Ma, apparently lacking any Grenvillian Pb-loss in an area where other geochronological evidence indicates such effects to be severe.

In addition to the dated intrusions being granite (sensu stricto), it has been observed that large bodies are confined to the southern part of the eastern Grenville Province, with only minor felsic magmatism farther north (Gower 1996). Two departures from these generalizations are the Rigolet quartz diorite in the northern part of the eastern Grenville Province, dated to be 1489^{+2}_{-8} Ma (Corrigan et al. 2000), and a 1458 ± 5 Ma diorite in the Manicouagan Imbricate Zone, immediately west of the area addressed here (Indares et al. 1998). The Rigolet body is neither granitic nor minor, and examination of the geochronological data suggests that an alternative interpretation should, at least, be considered. The age reported is based on one concordant fraction, anchoring the lower intercept of a poorly correlated line that includes two of the other four analyses. The line projects to a 2653 ± 37 Ma upper intercept, which begs more thorough study because this date is currently unique in intrusive rocks in the eastern Grenville Province. Offering an alternative interpretation, we argue that the scattered data points reflect pre-Labradorian inheritance, Labradorian emplacement, and (or) Labradorian, Pinwarian, and Grenvillian Pb-loss events. The pre-Grenvillian inheritance need not be Archean, although it must be at least 1986 Ma, the Pb–Pb age of the oldest discordant fraction. As the Rigolet quartz diorite is morphologically similar to the 1670 Ma Earl Island and 1675 Ma Susan River quartz diorites, has an comparable tectonic setting, and is geographically between them, we suggest that it likely has a similar age. A feature of Labradorian calc-alkaline rocks is that they are commonly intruded by amphibolitized mafic dykes. Corrigan et al. (2000) reasoned that their apparent absence from the Rigolet quartz diorite favours a post-Labradorian age, but such dykes are indeed present (e.g., locality CG80-194). Note that we find no reason to reevaluate the geochronological data from the diorite in the Manicouagan Imbricate Zone, which, in any case, is located very much closer to where Pinwarian calc-alkaline rocks might be anticipated, especially in the context of Grenvillian thrusting.

The qualifier "dated" Pinwarian magmatism was included above since a broad spectrum of magmatic products, other than granite, are present (leaving aside the Rigolet quartz diorite). For example, granite in the Upper Paradise River intrusion is spatially associated with pyroxene-bearing monzonite, anorthosite, and gabbronoritic mafic rocks, collectively representing a typical AMCG (anorthosite–monzonite–charnockite–granite) association. Furthermore, based on mapping and dating on its northern and western flanks (James et al. 2000; Madore et al. 1997*a*, 1997*b*), it seems likely that the Petit Mecatina AMCG intrusion is Pinwarian, rather than having, as previously supposed, an Adirondian (see later in the text) age.

Pinwarian deformation and metamorphism

Apart from demonstrating widespread granitic magmatism, geochronological data provide evidence of metamorphism and deformation throughout the same time period. Early metamorphism is indicated by ca. 1500 Ma ages from the Pleasure Cove quartzite and "The Lodge" quartz monzonite gneiss, which are coeval with a 1498^{+9}_{-8} Ma high-grade event that produced the Disappointment Hill felsic gneiss in western Newfoundland. Slightly younger is a 1493^{+6}_{-5} Ma granitic body that truncates the principal deformation fabric in granulite gneisses in the Hart Jaune complex but was intruded prior to the peak of metamorphism at 1487^{+7}_{-4} Ma, as suggested by Gobeil et al. (1996) on the basis of three near-concordant monazite analyses from a 1510 Ma mafic pegmatite. Coeval effects are echoed in the Hawke River terrane by a 1490 Ma titanite overprint on a Labradorian granite and 1490 Ma zircon from pegmatitic interboudin material in the Mealy Mountains terrane. Of additional interest is a 1509^{+11}_{-12} Ma date from a granitic vein at Long Harbour, argued by Scott et al. (1993) to have been emplaced into an actively deforming shear zone. Gower et al. (1996), in the face of lack of evidence for Pinwarian deformation, suggested alternatively that the pre-granite-vein mylonitization could be Labradorian and the post-granite-vein deformation Grenvillian. While this remains a viable possibility to us, we acknowledge that independent evidence for ca. 1500 Ma de-

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formation now exists elsewhere to support the interpretation of Scott et al. (1993).

High-grade metamorphism in the Hart Jaune terrane occurred at 1467^{+5}_{-4} Ma, the age of a nebulitic pod of granitic material in mafic granulites, and pre-dated a 1466 ± 5 Ma pegmatite that truncates folded high-grade rocks (Scott and Hynes 1994*a*, *b*). These dates are consistent with ca. 1469 Ma pegmatitic material between amphibolite boudins in the Mealy Mountains terrane (Krogh et al. 1996), a 1471 ± 41 Ma titanite age from mafic volcanic rocks and a ca. 1470 Ma monazite age from pegmatitic leucogabbro, the latter two both in the Hawke River terrane (Kamo et al. 1996). Precision latitude allows a metamorphic age of 1450^{+15}_{-21} Ma from the West of Trout River Quartz monzonite to be near coeval.

Early Elsonian

The term Elsonian "orogeny" was originally introduced by Stockwell (1964) to describe an event characterized by AMCG magmatism and reworking of older gneisses. As pointed out by Emslie (1978), the Elsonian never attained status as an orogenic event and was better regarded in a magmatic context — as has been subsequent usage. We consider the Elsonian to extend from 1460 Ma to 1230 Ma, time limits that are very similar to those presciently identified by Emslie (1978), who had far fewer, and much less precise, geochronological data at his disposal. It is divided here into three parts; early Elsonian between 1460 and 1350 Ma, middle Elsonian between 1350 and 1290 Ma, and late Elsonian between 1290 and 1230 Ma (Fig. 2).

Anorthositic and related magmatism north of the Grenville Province

Representatives of early Elsonian activity are all AMCG suites, namely the Michikamau, Harp Lake, and Mistastin complexes (Fig. 6). All are well north of the Grenville front, but their formation must be tectonically linked to similar-aged events in the Grenville Province, and it cannot be a coincidence that emplacement of these intrusions began at about 1460 Ma, coeval with the termination of Pinwarian magmatism.

The Michikamau complex is the oldest, based on crystallization ages of 1459 ± 2 Ma from the Michikamats megacrystic monzogranite (Krogh 1993), ca. 1460 and 1478 Ma (both Pb–Pb) from the Michikamau adamellite (Krogh and Davis 1973; Wanless, personal communication cited by Wanless and Loveridge 1978, respectively), and a 1469 ± 2 Ma metamorphic age (Krogh 1993) from a pre-Michikamau tonalite dyke that was hornfelsed during its emplacement of the Michikamau intrusion. A lower intercept date of 1456^{+91}_{-74} Ma (Krogh 1986) from an Archean tonalite east of the Michikamau complex must relate to the same event, but the outcrop is too far from surface exposures of either the Michikamau or Harp Lake bodies to be considered a product of contact metamorphism. The age of the Harp Lake intrusion is based on Pb–Pb dates of 1448 Ma from the Arc Lake adamellite (Krogh and Davis 1973) and 1426 Ma from the Snegamook granite (details unpublished; Wanless, personal communication cited by Emslie 1980). The only U–Pb data reported on the Mistastin suite is an unpublished 1420 Ma date mentioned by Emslie and Stirling (1993).

Other data having possible bearing on this period of geological activity north of the Grenville front are U–Pb metamorphic ages, involving long extrapolations to lower intercepts, of 1434 ± 4 Ma from the Akuliakatak Bay amphibolite dyke (Ryan et al. 1991) and 1410 ± 50 Ma on the Dumbell Island mafic gneiss inclusion (Connelly and Ryan 1996). Both dated rocks have early Paleoproterozoic or older protoliths, and the interpretation most in keeping with present geological information is that the metamorphism was related to emplacement of the 1350–1290 Ma Nain Plutonic Suite, explaining the apparently pre-1350 Ma ages as a result of incorporation of a small inherited component (Connelly and Ryan 1996).

Mafic magmatism in the northern Grenville Province: Michael and Shabogamo gabbros

Mafic magmatism during the early Elsonian is represented by a swarm of roughly east-northeast-trending, sheet-like bodies (mostly dykes) termed the Michael gabbro in eastern Labrador and the Shabogamo gabbro in western Labrador (Fig. 6). Collectively, the two suites form a belt about 700 km long and roughly 50 km wide that lies close to and south of the Grenville front, except in western Labrador where some intrusions occur north of the front. The Shabogamo and Michael gabbros are both tholeiitic, subalkaline transitional to mildly alkaline basaltic rocks. Based on whole-rock geochemical data, Gower et al. (1990*a*) concluded that they belong to a single swarm, an interpretation now substantiated by U–Pb geochronological data. Using Sr and Nd isotopic data Emslie et al. (1997) further concluded that the two suites had comparable evolutionary histories from a subcontinental lithospheric mantle source.

The most precise dates available are 1426 ± 6 Ma for the Michael gabbro (Schärer et al. 1986) and 1445 ± 4 Ma (Krogh 1993) for the Mount Fyne gabbro, a local name for the Shabogamo gabbro in the Churchill Falls terrane. Three other, less rigorous, U–Pb dates are 1459^{+23}_{-22} Ma and 1452^{+15}_{-13} Ma for the Shabogamo gabbro (Connelly and Heaman 1993; Connelly et al. 1995), and 1472^{+27}_{-21} Ma (Corrigan et al. 2000) for a gabbro near Rigolet. Aside from precision issues and the fact that both the 1459 and 1472 Ma regressions excluded analyses, caution is required in assigning the 1472 Ma sample to the Michael gabbro. Specifically, the whole-rock geochemical data of Tulk (1996) for the 1472 Ma sample has high Al_2O_3, CaO, and Na_2O and anomalously low Fe_2O_3t, MgO, and MnO — even compared to those Michael gabbros enriched in plagioclase and depleted in ferromagnesian minerals. The most concordant results of 1445 Ma and 1426 Ma for the Shabogamo and Michael gabbros, respectively, remain the most reliable guides to times of emplacement of this suite.

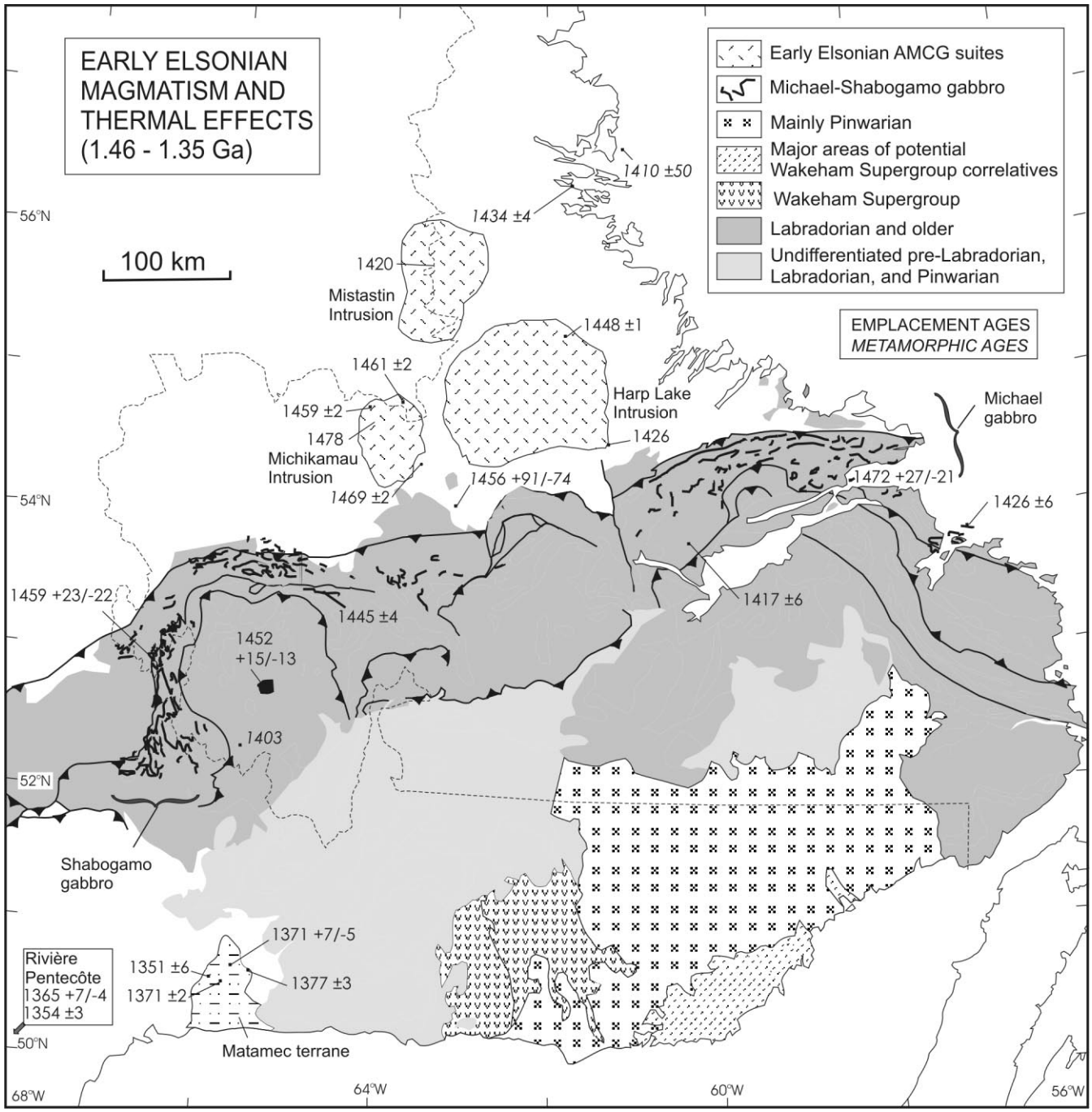
Quiescence between 1420 Ma and 1350 Ma (excluding the Matamec complex)

Evidence is currently lacking for significant geological activity between 1420 Ma and 1350 Ma throughout the whole of the eastern Grenville Province, excluding the Matamec complex in the extreme southwest part of the region. Apart from the Matamec terrane (see next text section), only two U–Pb dates fall within this 70 million year period.

The most tangible of the two is a zircon age of 1417 ± 5 Ma for the Mokami Hill quartz monzonite in the Lake Melville terrane (Gower and Kamo 1997). It should be noted that the regression producing this age excluded one fraction having a

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Fig. 6. Distribution of early Elsonian igneous rocks and U–Pb ages. AMCG, anorthosite–monzonite–charnockite–granite.



Pb–Pb age of 1422 Ma (1.1% discordant) and that the nominal lower intercept of 770 Ma is somewhat younger than a geologically sensible and more typical Grenvillian Pb-loss age of about 1000 Ma. A different treatment of the data could produce a marginally older age that would be closer to that of the early Elsonian AMCG suites north of the Grenville front, with which the Mokami Hill quartz monzonite is most logically grouped. The intrusion forms a distinctive, steep-sided hill rising over 400 m above its surrounding Labradorian gneisses. One might suppose that if other similar bodies existed in the eastern Grenville Province they would be equally obvious topographically.

The other date is from a late-Labradorian (1606 Ma) neosome from the Lac Joseph terrane. An age of 1403 Ma was obtained (Connelly and Heaman 1993), based on two concordant monazite analyses and excluding a third that contained a small component of inheritance. As the authors noted, this date does not correlate with any known metamorphic or structural elements in the area.

Matamec complex

The Matamec complex is a term introduced by Gobeil et al. (1999) for an assemblage dominated by ophitic to granoblastic gabbronoritic rocks associated with K-feldspar megacrystic mangerite and granite, and including later intrusions of gabbronorite and norite. A sheet of mangerite and a layer

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of augen granitic gneiss have yielded identical ages of 1371^{+7}_{-5} Ma and 1371 ± 2 Ma, respectively, (Fig. 6). The Lac Élinore porphyritic granite from a site farther east has an age of 1377 ± 3 Ma. As these rocks intrude or contain xenoliths of gabbronorite, an older age is indicated for the gabbronorite, although magma-mingling textures suggest great time differences are unlikely. The mineralized 1351 ± 6 Ma Lac Volant gabbronoritic dyke truncates granulite-facies fabrics in its 1371 Ma Matamec complex host, providing evidence of a significant deformational and metamorphic event between the two. These results were obtained from thrust-bound sheets that structurally overlie the surrounding gneisses, implying that the geological history established here need not be directly related to rest of the eastern Grenville Province (see later in the text).

Middle Elsonian

Middle Elsonian (1350–1290 Ma) events can be considered in two groups, namely (*i*) the Nain Plutonic Suite and (*ii*) coeval felsic rocks within the Grenville Province (Fig. 7).

Nain Plutonic Suite

Middle Elsonian time boundaries equate with the time span for emplacement of the AMCG Nain Plutonic Suite, from 1351 ± 3 Ma for the Hare Hill monzonite (Connelly 1993) to 1292 ± 4 Ma for a granite south of Kogaluk Bay (Ryan et al. 1991). Granulite-facies 1328–1316 Ma deformed tholeiitic dykes have been reported from the Nain Plutonic Suite by Connelly and Ryan (1999), although it is yet to be resolved whether these are regional in extent or confined to it. It is worth mentioning that, although there seems to be a dearth of geological activity before the emplacement of the Nain Plutonic Suite (70 million years, since the end of Mistastin intrusive activity), the same is not true for the period following the Nain Plutonic Suite, which merges without time break into the Flowers River Igneous Suite (see later in the text). Numerous U–Pb dates establish the time of emplacement of various intrusions in the Nain Plutonic Suite and thermal metamorphism of its Archean and Paleoproterozoic envelope, although analytical details of many determinations remain unpublished. They collectively indicate more-or-less continuous emplacement of anorthosite, norite–leuconorite, ferrodiorite, monzonite, and granite throughout this period, and a general west-to-east decrease in age.

Other activity occurred in the same region. Farther south, the Pants Lake Intrusion, which, like the Nain Plutonic Suite, has been a recent target for Ni–Cu–Co mineral exploration, has been dated to be 1322 ± 2 Ma (Smith et al. 1999). West of the Nain Plutonic Suite, the Napeu Kainiut quartz monzonite, into which the 1240 Ma Strange Lake complex was emplaced, is compared petrographically by Miller et al. (1997) to the ca. 1420 Ma Mistasin Intrusion and the 1322 Ma Makhavinekh Lake pluton, but no geochronological data are available to assign it specifically.

Felsic rocks in the Grenville Province coeval with the Nain Plutonic Suite

In contrast to the huge volumes of magma that produced the Nain Plutonic Suite, igneous activity within the Grenville Province during the middle Elsonian was sporadic and minor (Fig. 7). In central Labrador, close to the Grenville front, the spatially, temporally and genetically related Red Wine Intrusive Suite (Thomas 1983) and Letitia Lake Group (Thomas 1981) were emplaced. The Red Wine Intrusive Suite comprises granite, quartz syenite, nepheline syenite, malignite, and ijolite plutons (formerly designated as the Red Wine alkaline complex and the Arc Lake intrusive suite) and dykes that intrude the Letitia Lake Group. The latter is a supracrustal sequence that includes peralkaline and non-peralkaline quartz–feldspar porphyry, rhyolite and trachyte, and related volcanogenic sediments, including ignimbritic tuffs, muscovite–sericite schists, and volcanogenically derived sediments (Hill and Miller 1990). A peralkaline granite belonging to the Red Wine Intrusive Suite has an age of 1337^{+10}_{-7} Ma and a quartz–feldspar porphyry sill associated with the Letitia Lake Group an age of 1327^{+4}_{-2} Ma (Fryer 1983). The latter date must be regarded as preliminary as it is based on only one zircon fraction, albeit almost concordant.

Farther south in the Grenville Province, a monzonite recorded as underlying rhyolite of the Aguanus River Group was reported as having an age of 1322 ± 7 Ma (Martignole et al. 1994, citing Martignole et al. 1992). The age is presumably based on the same data set as that referred to by Bourne (1991, citing Desjardins et al. 1987), who mentioned a date of 1322 ± 3 Ma for a hypabyssal intrusion emplaced into the Aguanus River Group. The differing perspective of these two sources is important as Martignole et al. (1994) observed that the monzonite is intruded by gabbro dykes that, seemingly, are absent from the overlying rhyolites (dated to be 1271 Ma, see later in the text). More recent geochronological data indicates that at least part of the Wakeham Supergroup must predate 1500 Ma (see earlier in the text), requiring reinterpretation of the rhyolite as an intrusive rather than extrusive body. Regardless of these uncertain time relationships, the 1322 Ma date indicates that magmatic activity, coeval with the Nain Plutonic Suite, occurred well into the interior of the eastern Grenville Province.

There are two remaining reliably dated rocks in this time period within the eastern Grenville province. The first is the 1296^{+13}_{-12} Ma Upper North River granite in the Lake Melville terrane (Schärer et al. 1986). The age is anomalous for the district, but there seems no reason to challenge the data, especially as the rock is lithologically distinct from its surroundings, including hosting minor Mo and U mineralization. The second is a 1300^{+8}_{-5} Ma leucogranite from the Lelukuau domain (Indares et al. 1998), situated immediately west of the area addressed in this review.

Late Elsonian

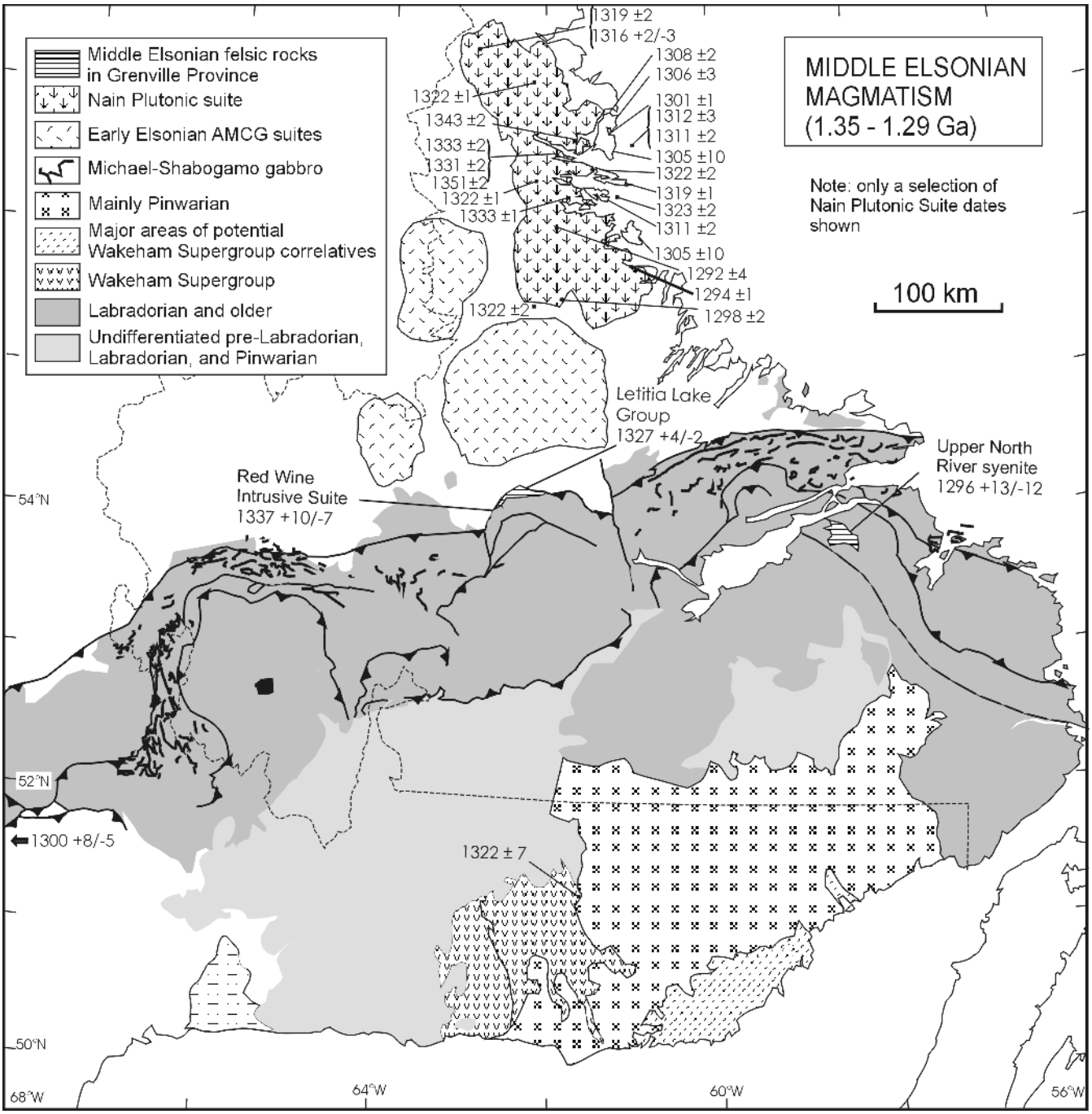
Late Elsonian magmatism can be spatially generalized as comprising felsic magmatism north and south of a central region of mafic magmatism, all occurring more-or-less concurrently (Fig. 8).

Felsic magmatism north of the Grenville Province

The earliest late Elsonian magmatism north of the Grenville Province was the formation of the Flowers River Igneous Suite, which comprises concentric peralkaline granites, surrounding a circular area of subaerial felsic volcanic rocks (Hill and Miller 1990; Hill 1991). These rocks are spatially and temporally continuous with the Nain Plutonic Suite, so it becomes a moot point whether introducing a time boundary

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Fig. 7. Distribution of middle Elsonian igneous rocks and U–Pb ages. AMCG, anorthosite–monzonite–charnockite–granite.

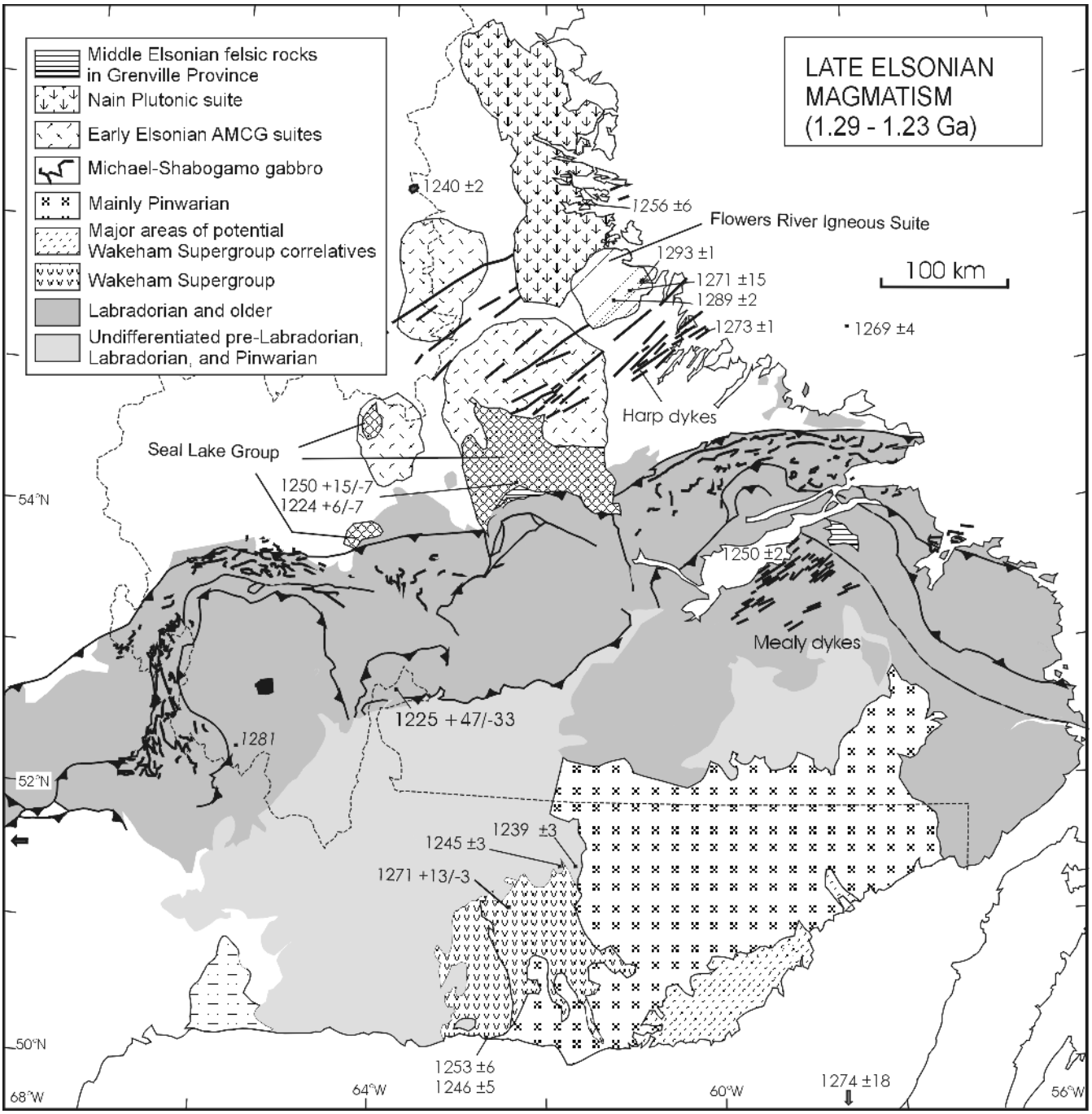


between them is justified. The rationale adopted here is that the Flowers River Igneous Suite introduces a contrasting type of magmatism following substantial uplift, presumed to reflect changes in deep crustal and mantle processes. Furthermore, by the end of the late Elsonian, conditions had radically changed from those at its outset, so we believe the time division can be defended. Time of emplacement of the Flowers River Igneous Suite is constrained by zircon dates of 1293 ± 1 Ma from a monzonite (Brooks 1983), 1289 ± 2 Ma from a rhyolite (Krogh 1993), and 1271 ± 15 Ma from a peralkaline granite (Brooks 1982). A zircon age of 1269 ± 4 Ma, based on chips of (?)anorthosite from an offshore well has been included with this group (Wasteneys et al. 1996).

Petrologically similar to the Flowers River Igneous Suite, but farther to the northwest is the Strange Lake peralkaline complex, an association of fine- and medium-grained, porphyritic, pegmatitic, and exotic-bearing granitoid rocks containing an abundant and varied collection of rare-metal-bearing minerals. A sample of fine-grained, exotic-poor granite yielded a concordant age of 1240 ± 2 Ma (Miller et al. 1997). The 1240 Ma age dates one of the oldest units recognized in the complex.

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Fig. 8. Distribution of late Elsonian igneous and supracrustal rocks and U–Pb ages. AMCG, anorthosite–monzonite–charnockite–granite.



Geothermobarometric conditions for emplacement of the Nain Plutonic Suite imply that between 9 and 15 km of material was removed prior to deposition of felsic volcanic rocks of the Flowers River Igneous Suite.

Mafic magmatism north and south of Grenville front

Mafic magmatism during the late Elsonian is represented, from north to south, by the Nutak, Nain, and Harp dykes (possibly including the Schefferville dykes); the Seal Lake Group; and the Mealy dykes (Fig. 8).

Nutak dykes

The Nutak dykes are generally north- to north-northwest-trending intrusions north of Nain and east-southeast-trending dykes within the Nain Province and are described as olivine gabbro, containing up to 70% plagioclase (Ermanovics et al. 1989; Ermanovics, personal communication to Gower et al. 1990*a*; Cadman et al. 1993). An unpublished U–Pb age of ca. 1268 Ma from an east-southeast-trending dyke is mentioned by Cadman et al. (1993, citing Roddick, personal communication).

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Nain dykes

The Nain dykes (a term informally introduced by Gower et al. 1990*b*) intrude the Nain Plutonic Suite and have been described in detail by Wiebe (1985). Wiebe divided the Nain dykes into two compositionally distinct groups termed low P_2O_5 (LP) and high P_2O_5 (HP) types. The LP dykes are east-northeast-trending, in contrast to the HP dykes, which define an arcuate trend form northwest in the south through north to northeast in the north. The relative age of these two dyke suites is unknown as no crosscutting relationships were observed. Carlson et al. (1993) identified the LP dykes as high-Fe tholeiitic to transitional alkalic composition, similar to some of the magmas involved in the formation of the Nain Plutonic Suite, and the HP dykes as having intraplate, oceanic alkalic basaltic affinities. The two dyke types are not related by crustal contamination, fractional crystallization, or different degrees of melting of a common source.

Harp dykes

The Harp dykes are a suite of east-northeast-trending intrusions emplaced into the Harp Lake complex (Emslie 1980) but extending beyond it into the surrounding country rocks of both the Churchill and Nain Provinces. Cadman et al. (1993) subdivide the Harp dykes into three groups, termed the Harp CO, UT, and PT dykes. The Harp CO dykes (after Comma Island) are thin, north- or northeast-trending, and characterized by negative phosphorus (P) anomalies on spidergrams. The Harp UT dykes (after Umiatoriak Island) are thicker, northeast- or east-northeast-trending and characterized by high Ba. The Harp PT dykes (after Pitsiutatsitikulluk Island) have similar trends and chemical characteristics to the UT dykes, but have negative phosphorus anomalies. A zircon age of 1273 ± 1 Ma from a Harp UT dyke was reported by Cadman et al. (1993).

It is tempting to correlate the east-northeast-trending Harp dykes (UT and PT) with the similarly trending Nain LP dykes and Nutak dykes, and the north to east-northeast-trending Harp CO dykes with the similarly trending Nain HP dykes and some Nutak dykes (cf. Gower et al. 1990*a*; Cadman et al. 1993), but such correlations are open to challenge based on chemical and magnetic expression differences. Cadman et al. (1993) suggested that several swarms might be present.

The Schefferville dykes, situated farther west, have not been precisely dated, but paleomagnetic and K–Ar radiometric data permit correlation with the Harp dykes (cf. Buchan and Halls 1990). Note, however, that their north-northeasterly trend is at variance with the prevailing orientation of most Harp dykes.

Seal Lake Group

The Seal Lake Group (Fahrig 1959) is a sequence of sedimentary rocks, volcanic rocks and associated sills that straddle the northern margin of the Grenville Province. Outliers of similar rocks north and west of the main depocentre, and within the northern Grenville Province to the south, suggest that Seal Lake Group may have been much more extensive than presently preserved (Hill and Miller 1990; Wardle et al. 1990). The sequence has been variously estimated to be between 5000 to 14 000 m thick, and its stratigraphy has been described by Ryan (1984 and references therein). The basal part of the Seal Lake Group is characterized by arkose and conglomerate, which are overlain by a thick sequence of mafic flows. The central part of the Group is dominated by quartzite, siltstone, shale, and limestone that were intruded by numerous diabase and gabbro sills (Naskaupi sills). These rocks grade up into shale, siltstone, quartzite, and subordinate mafic flows and sills. Flows within the Seal Lake Group are amygdaloidal, mostly subaerial basalt, but rare pillows and hyaloclastite breccias occur near the base of the sequence. The Naskaupi sills, intruding the central and upper parts of the Seal Lake Group, range from 30 to 200 m thick and are fractionated, having ophitic gabbro in their lower parts and pegmatitic zones at high levels. Conglomerate, preserved on a low-lying part of the Harp Lake Complex and believed correlative with the Seal Lake Group, contains clasts of diabase similar to that of the Harp dykes (Irving et al. 1977). The maximum age of the Seal Lake Group is constrained by the presence of clasts of quartz–feldspar porphyry and felsic volcanic rocks in basal conglomerates of the Seal Lake Group, derived from the unconformably underlying 1327 Ma Letitia Lake Group (Martin 1975; Thomas 1981).

The age of the Seal Lake Group is constrained by zircon–baddeleyite emplacement dates of 1250^{+14}_{-7} Ma and 1224^{+6}_{-5} Ma (Romer et al. 1995) from two different gabbro sills in the Wuchusk Lake Formation, hence providing a minimum age for the supracrustal rocks they intrude.

Mealy dykes

The Mealy dykes form an east-northeast-trending swarm of olivine gabbro and diabase intrusions hosted by the 1640 Ma Mealy Mountain Intrusive Suite and the immediately surrounding country rocks (Emslie 1976; Emslie et al. 1984). Time of emplacement is given by a zircon age of 1250 ± 2 Ma (Hamilton and Emslie 1997).

Felsic rocks within the Grenville Province

The most detailed information regarding late Elsonian events in the eastern Grenville Province comes from the Wakeham Supergroup (Fig. 8). A porphyritic rhyolite from the Aguanus Group, mapped by Bourne et al. (1977) as extrusive, is the oldest dated rock, having a zircon age of 1271^{+13}_{-3} Ma (Loveridge 1986). Its volcanic origin was endorsed by Martignole et al. (1994), but this interpretation has been thrown into doubt by 1510 to 1495 Ma porphyries that intrude the Wakeham Supergoup (see earlier in the text). If it is insisted that supracrustal rocks of only one age are present, then the rhyolite must be intrusive. In the lower part of the Davy Group, two fractions from a K-feldspar porphyry yielded concordant zircon ages of 1253 ± 6 Ma and 1246 ± 5 Ma, and another fraction gave a 1.9% discordant Pb–Pb age of 1265 Ma (Clark and Machado 1995). The porphyry was considered volcanic by the authors. Added to these results are ages of 1245 ± 3 Ma and 1239 ± 3 Ma from two samples of Natashquan River granite (Martignole et al. 1994), now included as part of Madore et al.'s (1997*a*, 1997*b*) Olomane Suite.

Two dates stand out in stark contrast to the felsic magmatism representative of the remainder of this period, but perhaps neither are greatly significant. The first is a 1281 Ma (titanite) date from a 1648 Ma mafic migmatitic gneiss in the Lac Joseph terrane (Connelly and Heaman 1993) and the second is a 1274 ± 18 Ma (zircon) result from amphibolite regarded

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as a border phase of the Steel Mountain anorthosite complex in the Southern Long Range Mountains of Newfoundland (Currie and van Berkel 1992). The former was interpreted to reflect a hydrous amphibolite-facies retrogression in the area, but no explanation for the latter was offered by the authors.

Tectonic model

Post-Labradorian – pre-Pinwarian (1600–1520 Ma)

The period between the end of the Labradorian orogeny and the start of the Pinwarian activity has been recognized for some time as one of quiescence throughout Laurentia (Gower et al. 1990*b*). That this period in the eastern Grenville Province might be linked with a passive continental margin was suggested by Gower (1996), who hypothesized that large tracts of undated metasedimentary gneiss flanking the north shore of the Gulf of St. Lawrence provide evidence of its former existence. These rocks remain undated, but parts of the Wakeham Supergroup, formerly thought to have a ca. 1270 Ma age, have since been determined to postdate Labradorian events and predate Pinwarian orogenesis. Appropriately, they also contain material derived from pre-Pinwarian Laurentia. The Wakeham Supergroup, along with other previously and recently mapped rocks of similar lithological character in the southern part of the eastern Grenville Province that may be correlative, therefore, provides some formerly lacking evidence for a passive margin.

Pinwarian (1520–1460 Ma)

Tectonic models for dominantly felsic magmatism, with associated deformation and metamorphism between 1520 and 1460 Ma in the Grenville Province (now termed Pinwarian) have mostly oscillated between some form of within-plate extensional setting to a continental-margin arc. The balance of evidence has gradually shifted over time from the former to the latter. For the southwest Grenville Province, perhaps the first to suggest a subduction-related environment during geon 14 were van Breemen and Davidson (1988), who challenged then-prevailing concepts (applied to coeval felsic rocks in the mid-west U.S.A.) that the magmatism was the product of crustal melting in an extensional environment (Bickford et al. 1986; Windley 1989). In their contribution, van Breemen and Davidson (1988) drew parallels with mid-Tertiary igneous activity along the west coast of North America, suggesting that the magmatism could be related to subduction-zone steepening.

At that time, magmatism of similar age and type was unknown in the eastern Grenville Province. It was first reported in preliminary form by Tucker and Gower (1990). Subsequently coeval rocks were identified by Currie et al. (1992) in western Newfoundland, and then the subject was addressed in more detail by Tucker and Gower (1994) and Gower and Tucker (1994). The name "Pinwarian" was proposed by Tucker and Gower (1994) for a regionally extensive magmatic event between 1500 and 1470 Ma. This is now extended to between 1520 and 1460 Ma on the basis of later geochronological studies. The event is now regarded as orogenic because accompanying deformational and metamorphic activity have been identified. That Pinwarian magmatism in the eastern Grenville Province might be related to a continental-margin arc was first proposed by Gower and Tucker (1994), influenced by Sm–Nd data from the midwestern U.S.A. (Bowring et al. 1992) and from the southern part of the central Grenville Province (Dickin and Higgins 1992), both suggesting the existence of ca. 1.5 Ga juvenile crust. Gower (1996; echoed by Rivers 1997; and Rivers and Corrigan 2000) further advocated Pinwarian orogenesis to be the product of a post-Labradorian Laurentian continental-margin arc (i.e., over a northward-dipping subduction zone). Gower (1996) emphasized that the bulk of the arc must have been south of then-investigated areas, as there is clear evidence that Labradorian crust extended to southernmost Labrador. Since then, Pinwarian magmatic products have been increasingly identified in southernmost Labrador and adjacent eastern Quebec (Heaman et al. 1996; Madore et al. 1997*a*, 1997*b*; Verpaelst et al. 1997*a*, 1997*b*; Wasteneys et al. 1997; James et al. 2000, 2001).

Despite mounting evidence in favour of a continental-arc tectonic setting, it should be kept in mind that almost all dated rocks are K-rich granites or AMCG-type lithologies, rather than typical calc-alkaline products. This could mean that they were generated inboard of the main arc.

Early to mid Elsonian (1460–1290 Ma)

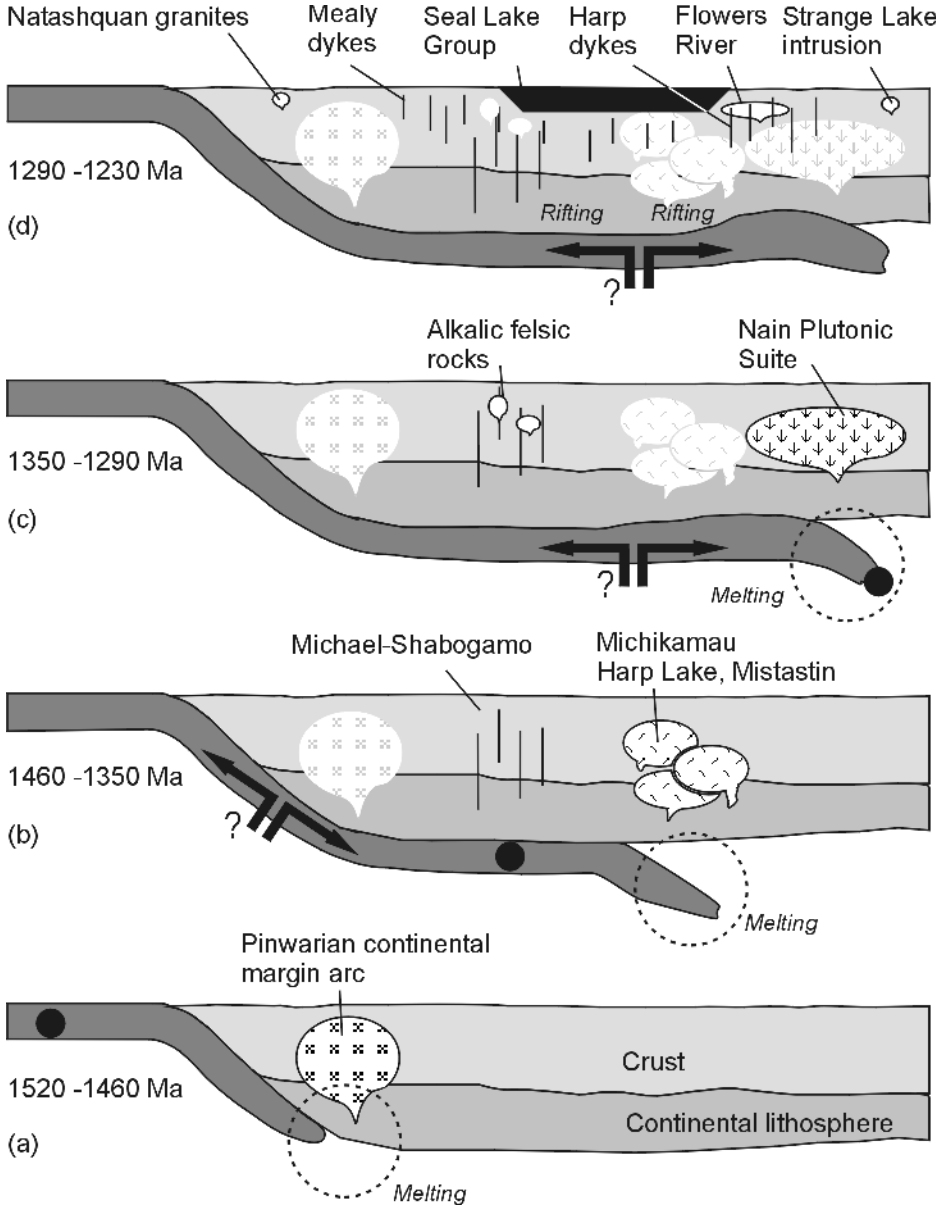
On the basis of present data, Pinwarian magmatism ceased by 1460 Ma, and the focus of activity shifted to the northern fringe of the eastern Grenville Province and farther north. It is worth emphasizing that calc-alkaline magmatism has not been recognized in the eastern Grenville Province during Elsonian or later times. This means that models for a continental-margin arc over this extended period must account for the absence of appropriate rock types anywhere in the region.

During the early Elsonian between 1460 and 1420 Ma, two contrasting, but coeval and undoubtedly genetically related, events occurred. The Michael–Shabogamo gabbros were emplaced in the northern Grenville Province between 1459 and 1426 Ma, and the Michikamau, Harp and Mistastin AMCG suites were emplaced north of the Grenville front. Present data are now adequate to suggest a northward younging of these three AMCG suites. The Michikamau suite was most likely emplaced at 1460 Ma, the Harp Lake suite between 1448 and 1426 Ma, and the Mistastin suite at 1420 Ma. An exception to this pattern is the volumetrically minor 1417 Ma Mokami Hill intrusion within the Groswater Bay terrane.

Following a period of apparent quiescence from 1420 Ma to 1350 Ma (except the Matamec terrane — see later in the text), the pattern of northward-advancing AMCG magmatism continued, as demonstrated by the emplacement of the Nain Plutonic Suite between 1350 and 1290 Ma. There is indication of a west-to-east younging within the Nain Plutonic Suite. Given that the older Mistastin intrusion lies to the west, a systematic younging trend is followed, although this may be a second-order, west-to-east trend. Overall, the northward progression appears to have taken place in a zig-zag fashion (perhaps governed in part by space availability?). Unlike the early Elsonian, no mafic dyke swarms comparable to the Michael–Shabogamo gabbros have been identified. Nevertheless, it is established that mafic dykes were emplaced coeval with the Nain Plutonic Suite (Connelly and Ryan 1999), thus, as many periods of (undated) mafic dyke injection are known (Cadman et al. 1999), it is possible that

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Fig. 9. Tectonic model for the evolution of the eastern Grenville Province during Pinwarian and Elsonian periods.



an analogue for the Michael–Shabogamo swarm will be identified.

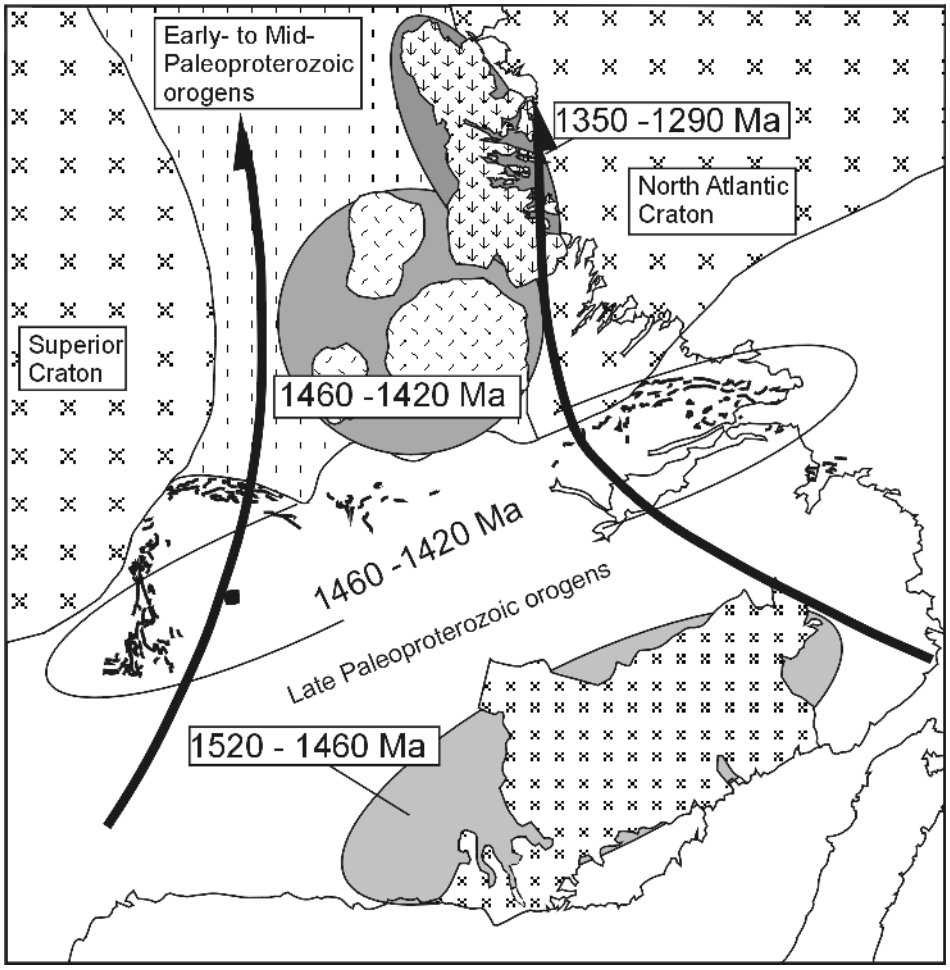
An additional point that we consider relevant is that the shape of bodies changed both in time and space. The southerly 1510–460 Ma Pinwarian intrusions collectively have an elongate east–west trend; the 1460–410 Ma early Elsonian intrusions in the middle are rounded, and the northern 1350–290 Ma Nain Plutonic Suite has a north–south trend. These trends are mimicked in part by associated mafic dykes (see later in the text).

Applied simplistically, we do not find any of the three most obvious models (back-arc setting, an intra-continental – failed rift or a mantle plume) very compelling. The northward decrease in ages is especially difficult to explain in either a static intra-continental or back-arc rift environment. Furthermore, the back-arc model demands an arc–trench gap of at least 800 km (excluding structural telescoping during Grenvillian orogenesis) if the Nain Plutonic Suite is to be accommodated — which it must, as to do otherwise demands special pleading. A mantle plume adequately explains both the south-to-north decrease in ages and distance from the continental margin, but has difficulties in explaining the south-to-north, oblate-to-prolate changes in intrusive shapes. The 700 km-long swarm of Michael–Shabogamo gabbros also seems an unlikely product of a mantle plume.

Keeping in mind that we advocate a continental-margin arc for the period preceding 1460 Ma (Fig. 9*a*), we feel compelled to consider subduction-related models first. Furthermore, we feel obliged to accommodate absence of evidence for any continental-margin arc after 1460 Ma. Lacking indication for island-arc or continent accretion around 1460 Ma, two solutions come to mind. The first is some far-field effect, such as a change in tectonic regime on the other side of Laurentia. This is certainly a possibility as Belt

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Fig. 10. Change from oblate to prolate shape in magmatic products and decrease in their age progressing northward; related to "funnelled flat subduction" (see Fig. 11).



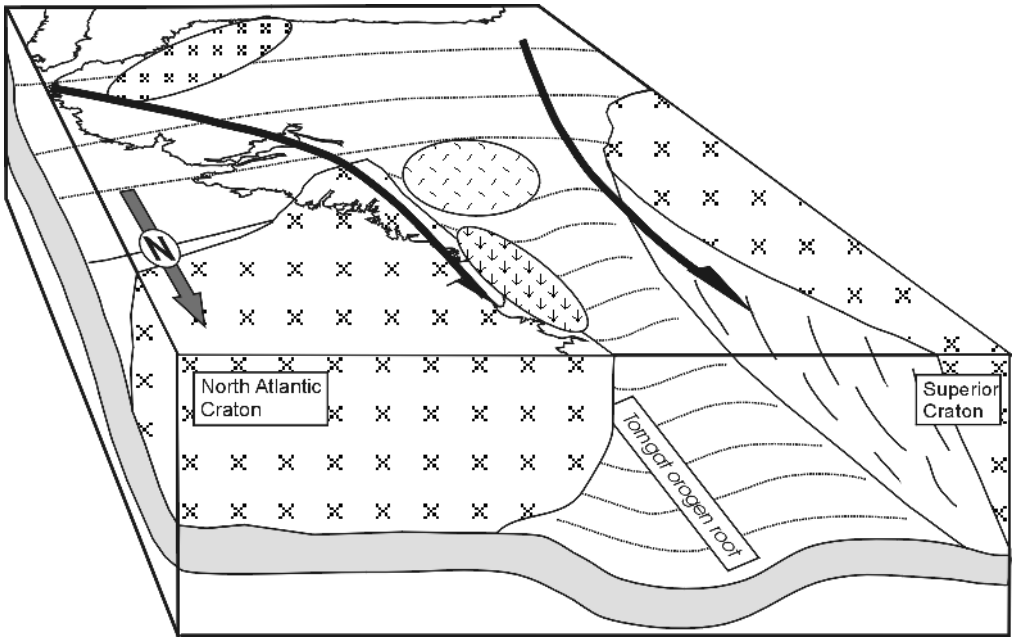
Supergroup deposition started in response to rifting on the western side of North America at this time. The second is a local solution that involved a change from regular oceanic crust being supplied to the subduction zone to an oceanic plateau or a spreading ridge. We favour a hypothesis whereby a spreading-ridge was overridden and flat subduction ensued (Figs. 9*b*, 9*c*). Gutscher et al. (2000*a*, 2000*b*) have presented models whereby, under flat subduction conditions, active magmatism at the continental margin ceases and oceanic crust penetrates several 100 kilometres inboard of the continental margin at the same depth. As the south-to-north decrease in ages is normal to the tectonic grain, we would argue that convergence was approximately normal to the post-Pinwarian, Laurentian continental margin. Gutscher et al. (2000*b*) appeal to the buoyancy of thickened oceanic crust, such as is found in oceanic plateaus, as a cause of flat subduction. We speculate that, applied to the eastern Grenville Province and region to the north, an overridden spreading-ridge might be required to provide the conditions needed to deliver mantle-derived melts and foster basal crustal melting, which have been consistently demanded by genetic models explaining AMCG suites. This is not to say that all magmatism was necessarily directly linked to the overridden spreading centre. The whole region underpinned by flat-subducted oceanic crust is one of tectonic instability and pockets of magmatism might be anticipated in any weak spot. In this way, magmatism that generated the Red Wine Intrusive Suite, Letitia Lake Group, Upper North River granite and monzonite intruding into the Aguanus Group could be accommodated.

This discussion has not yet addressed the south-to-north oblate-to-prolate transition (Fig. 10). We would argue that this is a consequence of a funnelling effect superimposed on a flat subduction regime (herein labelled "funnelled flat subduction"), a hypothesis that incorporates elements from both the flat-subduction model of Gutscher et al. (2000*a*, 2000*b*) and the mantle plume model of Funck and Louden (2000), the latter specifically for the Nain Plutonic Suite. As the leading edge of the flat subduction zone incrementally advanced north, it was forced below the north–south-trending roots of the Torngat orogen (Fig. 11). Such roots are postulated by Funck and Louden (2000) to exist between the Archean Superior and Nain cratons on the basis of seismic refraction data. As a result, melting was renewed along the leading edge of the shallowly subducted lower plate mafic crust and initiated above it. The funnelling may have been responsible for the crustal doming that took place near the end of the middle Elsonian, resulting in the erosion of 9–15 km of upper crust.

We acknowledge that a weakness of this model is that it does not account for 1380–350 Ma gabbronoritic to

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Fig. 11. Sketch showing funnelled flat subduction due to lower plate forced downward beneath Paleoproterozoic orogenic root (which is omitted to improve clarity).



monzonitic rocks in the Matamec terrane. They are almost certainly allochthonous, but accepting this only displaces the problem elsewhere. One pertinent observation is that these rocks occupy a time period unrepresented magmatically in other parts of the eastern Grenville Province. We speculate that the Matamec complex is linked to a distinct 1420–1350 Ma phase of activity concentrated in the south-central part of the Grenville Province (in Quebec). A nearby representative of this event is the Rivière Pentecôte intrusion (cf. Fig. 6), from which a zircon date of 1365^{+7}_{-4} Ma has been obtained from a quartz monzonite (Emslie and Hunt 1990) and 1354 ± 3 Ma from a norite (Machado and Martignole 1988).

Late Elsonian (1290–1230 Ma)

Two key features of late Elsonian times are (*i*) there was no real hiatus in magmatic activity defining the end of the middle Elsonian and the start of the late Elsonian, and (*ii*) northward younging of magmatism is no longer evident. It would seem axiomatic that the northward advance of flat-subducted oceanic crust must have a limit, after which further inboard incursion ceases. We suggest that this limit had been reached by 1290 Ma, and final delivery of heat from the mantle was centred under the present site of the Seal Lake Group, although affecting a large area of the surrounding crust. At this point, an archetypical back-arc basin could start to develop (Fig. 9*d*).

Emplacement of mafic magmas through the crust would have been greatly facilitated by the preceding crustal doming and consequent erosion. The crustal doming also offers an explanation for less-well-defined trends evident in late Elsonian mafic dyke swarms. In the southerly swarms, an east-northeast trend is dominant, as demonstrated by the Harp, Nain LP, and Mealy dykes. Farther north, the Nutak and Nain HP dykes have north to north-northwest trends, as would be expected as a result of changes in the principle extension direction in connection with funnelled flat subduction.

The alkalic felsic magmatism on the north and south flanks of the central area of dominantly mafic magmatism is no different from that associated with rifts elsewhere (for example, the Port Coldwell and other alkali complexes associated with the Mid-Continent Rift System). The Flowers River Igneous Suite, emplaced between 1293 Ma and 1269 Ma is regarded here as bridging the transition from northward-migrating funnelled flat subduction to stationary conditions (subduction shift to static rift), and other sporadic alkali-rich magmatism, such as the rhyolitic porphyries intruding the Wakeham Supergroup and the Strange Lake intrusion. The youngest mafic magmatism occurs in the Seal Lake Group, and along with coarse clastic, quartz-rich sediments would have developed into a full-fledged back-arc basin (probably accompanied by an active continental-margin arc farther south — Natashquan River granites?), if its development had not been arrested by Elzevirian orogenesis.

Part III. Elzevirian to Grenvillian (pressure-point orogenesis)

Coeval with Elzevirian orogenesis (1230–1180 Ma)

As a concept that identifies late geon-12 to early geon-11 orogenesis, the Elzevirian orogeny had its birth in the southwest Grenville Province (Moore and Thompson 1980). It is considered to have started there some time between 1250 and 1230 Ma and ended before 1180 Ma — exact time limits differing according to authorship (cf. Lumbers et al. 1990; Easton 1992; Carr et al. 2000). In tectonic terms, Elzevirian orogenesis represents the time of amalgamation of arc-related terranes in the Central Metasedimentary Belt and (or) their accretion to pre-Elzevirian Laurentia. The object here is not to enter the debate on the meaning of the Elzevirian orogeny

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in its type region, but instead to review and evaluate events that might have occurred during the same period in the eastern Grenville Province.

Right from the start it must be emphasized that evidence for an "Elzevirian" orogenic event in the eastern Grenville Province is meagre. In fact, there are no geochronological data from the eastern Grenville Province or peripheral regions that can be reliably considered to be Elzevirian, although three dates nominally within this time period require mention.

The first is the Lac Long Igneous suite, situated in the central part of southern Labrador, which consists of tonalite to quartz diorite gneiss, charnockitic granitoid rocks, and anorthosite. Four zircon fractions from a granulite-facies, tonalite – quartz gneiss plot in the middle of a chord having an upper intercept at 1225^{+47}_{-33} Ma and a lower intercept at 1008 ± 30 Ma (Krogh 1985; Thomas et al. 1986). Guided by zircon morphology, the older date was interpreted as approximating the age of the protolith and the younger date as being the time of Grenvillian metamorphism. Error limits allow a substantially pre-Elzevirian age, but, even if precision concerns could be disregarded, the nominal upper intercept dates emplacement rather than tectonism, so the result provides no evidence of orogenesis.

The second date is from the Baie du Nord anorthosite in the Tshenukutish terrane, immediately to the west of the area designated here as eastern Grenville Province. Based on five zircon fractions, upper and lower intercepts of 1214^{+49}_{-23} Ma and 1021^{+54}_{-69} Ma were obtained, although if the lower intercept is anchored at 1021 ± 5 Ma, the precision error on the upper intercept is reduced to 1214^{+31}_{-19} Ma (Indares et al. 1998). As the authors do not explain why the smaller error on the lower intercept is to be preferred, prudence demands adoption of the larger upper intercept error, hence caution in regarding this date as necessarily "Elzevirian." Again, emplacement rather than deformation or metamorphism is indicated. The low precision is unfortunate, as this date could have demonstrated that Elsonian conditions did not cease in the eastern Grenville at 1230 Ma and that much greater continuity exists with post-"Elzevirian" events than hitherto accepted.

The third date is from the Sailor Brook gneiss in Cape Breton, and obtained from a weakly layered dioritic to tonalite, weakly migmatitic rock. In this case, because of the combined effects of Pb-loss from two subsequent metamorphic events at 1035 Ma and 423 Ma, protolith formation cannot be established beyond a minimum age of 1217 Ma (Miller et al. 1996). It cannot be regarded as having "Elzevirian" relevance.

With doubts cast upon direct geochronological evidence for events that might be considered to fall within the 1230–1180 Ma period, it is necessary to resort to indirect approaches. The most reliable of these, in the case of deformation, means bracketing between two dated rocks at the same site, affected and unaffected by deformation, respectively. An approximation of this approach can be applied to the Wakeham Supergroup, inasmuch as the inferred-to-be-earlier Aguanus subdivision in the east is affected by two deformational events, whereas the Davy subdivision in the west is only affected by one. The minimum age of the Aguanus Group is 1239 Ma, based on the age of the Natashquan River granites (Olomane felsic suite) that intrude it, and the minimum age of the Davy Group is 1177 Ma, based on a date from a gabbro sill within it (see later in the text). As the sills are folded, the second deformation certainly postdates 1177 Ma. The emplacement of the Natashquan River granites at 1240 Ma is interpreted to have been associated with high-temperature (*T*) – low-pressure (*P*) metamorphism and the development of the Natashquan shear zone to have predated the Lillian mafic suite, which may have been emplaced at the same time as the gabbro sills in the Davy Group (see later in the text and Verpaelst et al. 1997*a*, 1997*b*). These inferences permit the deformation to be Elzevirian, but do not prove it.

Post-Elzevirian – pre-Grenvillian (Adirondian)

As used here, post-Elzevirian – pre-Grenvillian refers to the period between 1180 Ma and 1080 Ma. Activity was concentrated in the eastern Grenville Province during the first 60 million years, with apparent quiescence for up to 40 million years after ca. 1120 Ma. Gower et al. (1997*a*) informally adopted the term Adirondian for this period, as rocks of this age are well represented in the Adirondacks. This usage is continued here (cf. Fig. 12).

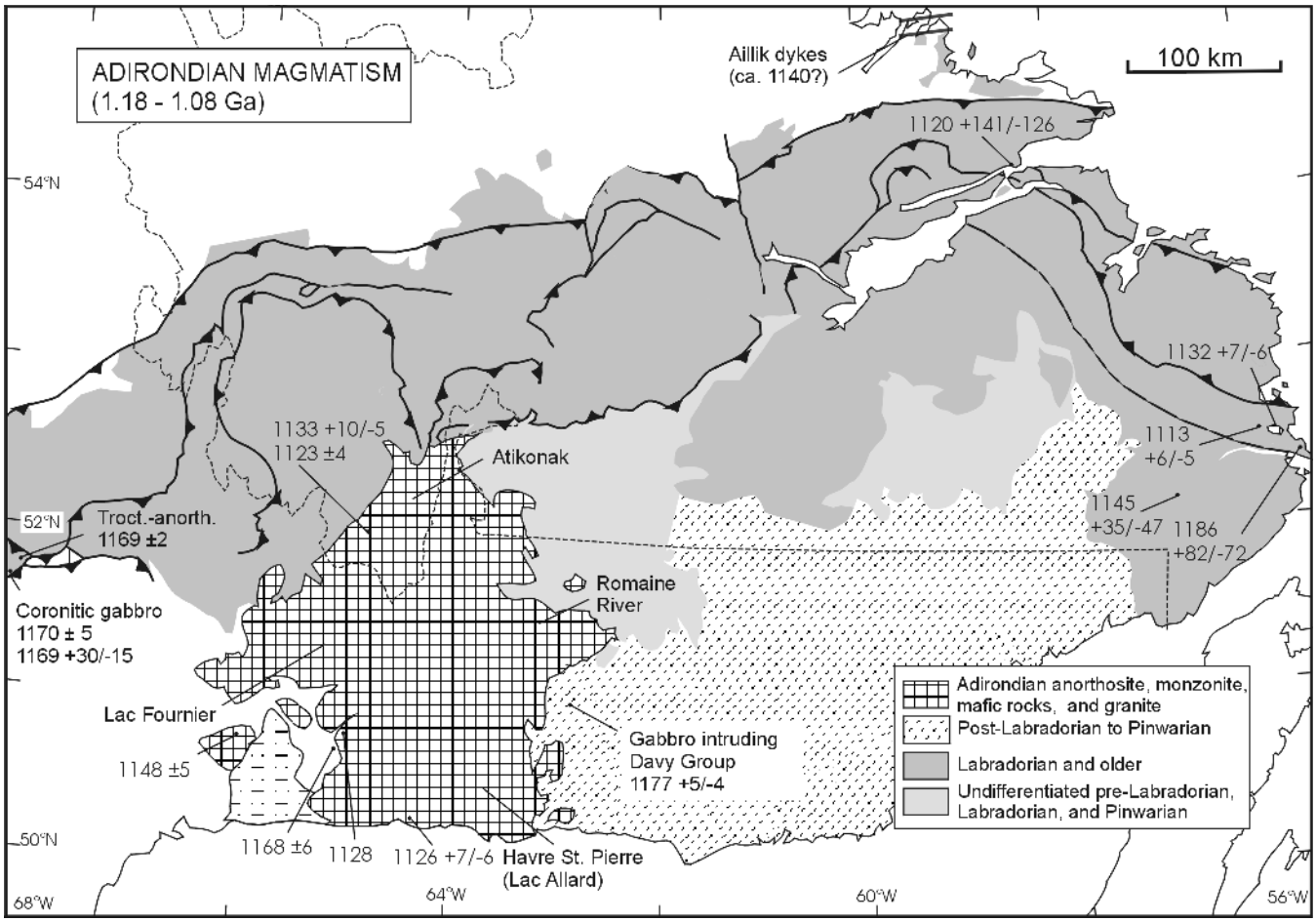
Adopting a 1180–1120 Ma active time span marginalizes three dates reported from granitic veins in eastern Labrador. The results are 1186^{+82}_{-72} Ma (Scott et al. 1993), 1120^{+141}_{-126} (Corrigan et al. 2000), and 1113^{+6}_{-5} (Scott et al. 1993). The first two are both based on imprecise lower intercepts from minor intrusions having Labradorian ages, and it would be unwise to assign much credence to the results. Similarly, Scott et al. (1993) express doubts that the 1113 Ma date necessarily denotes the emplacement age of the granitic vein investigated by them, suggesting that it might have been emplaced at 1078 Ma (based on concordant monazite from the same sample). Despite Scott et al.'s (1993) reservations, we favour a ca. 1113 Ma time of emplacement, arguing that the granitic vein is genetically part of the 1132 Ma Gilbert Bay pluton (with which it is in close proximity), but suggest that the two rocks might be closer in age than their nominal ages indicate. An additional "Adirondian" date of 1145^{+35}_{-47} Ma (Tucker and Gower 1994) can be rejected. Tucker and Gower (1994) advised caution in its use for reasons best appreciated by reading the original publication. Based on Pb-loss histories from other rocks in the region, Wasteneys et al. (1997) suggested a Labradorian protolith age for the "1145 Ma" granodiorite — a position fully endorsed here.

Thus filtered, the remaining ages define a clear pattern of mafic–anorthositic–monzogranitic (i.e., AMCG-type) magmatism confined to the southern, especially southwestern, part of the eastern Grenville Province (Fig. 12). One of the earliest manifestations of this event is the Robe-Noire mafic suite (Verpaelst et al. 1997*a*, 1997*b*) intruding the Davy subdivision of the Wakeham Supergroup and dated to be 1177^{+5}_{-4} Ma (Martignole et al. 1994). The Lillian mafic suite, dominated by east-trending dykes, and the Kapanien gabbro, both northeast of the Wakeham Supergroup, have not been dated but are correlated with the Robe-Noire mafic suite by Verpaelst et al. (1997*a*, 1997*b*) and Madore et al. (1999). That emplacement of mafic rocks at this time may have been widespread is supported by a 1170 ± 5 Ma gabbro (Cox et al. 1998) in the Manicouagan area.

Magmatism was not entirely mafic, as demonstrated by the 1169 ± 2 Ma (Scott and Hynes 1994*b*) Brian troctolite–anorthosite (and possibly other similar, but

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Fig. 12. Distribution of magmatic products during the Adirondian period. troct.-anorth., troctolite–anorthosite.



undated, intrusions described by Scowen et al. 1993) and 1168 ± 6 Ma from a small leucogranite pluton on the west flank of the Havre St. Pierre suite (Gobeil et al. 1999). In the Lac Nipisso area, a slightly younger porphyritic granite was emplaced at 1148 ± 5 Ma (Gobeil et al. 1999).

The two main massifs generated during this period were the Atikonak – Lac Fournier and Havre St. Pierre (Lac Allard) – Romaine River suites. The Atikonak River massif consists of leucotroctolite, troctolite, leuconorite, norite, and anorthosite associated with pyroxene monzonite, pyroxene quartz monzonite, and granite (Nunn et al. 1986). Dates of 1133^{+10}_{-5} Ma from a rapakivi granite (termed the Fleur-de-Mai granite by Connelly 1991) and 1123 ± 4 Ma from a pyroxene quartz monzonite were reported by Emslie and Hunt (1990). The 1123 Ma age is based a concordant single zircon, which was considered superior to a 6-point regression yielding upper and lower intercepts of 1652^{+31}_{-25} Ma and 1108^{+12}_{-13} Ma, respectively. The nominal 1652 Ma date is nevertheless useful in that it hints at the age of the source and (or) host rocks, in the absence of nearby direct geochronological data. The Havre St. Pierre – Romaine River suite, which hosts the world-class Lac Tio ilmenite deposit, has been dated to be 1128 Ma from anorthosite and 1126^{+7}_{-6} Ma from pyroxene quartz monzonite (Emslie and Hunt 1990). The 1132^{+7}_{-6} Ma Gilbert River pluton intruded during this time span, but Gower et al. (1991) warned that the age could not be considered rigorous as one analysis was excluded from the regression. In similar suites that have been more thoroughly dated elsewhere (e.g., the Nain Plutonic Suite and the Wiborg massif in Finland), it has been recognized that magmatism persisted over several tens of millions of years. Despite the apparent clustering of ages at ca. 1170 Ma and ca. 1130 Ma for these bodies in the eastern Grenville Province, it is possible that a similar pattern of extended emplacement, like that documented from the Nain Plutonic Suite will be established when more detailed geochronological data are available. It is worth emphasizing that all results date emplacement, with no indication of regional metamorphism.

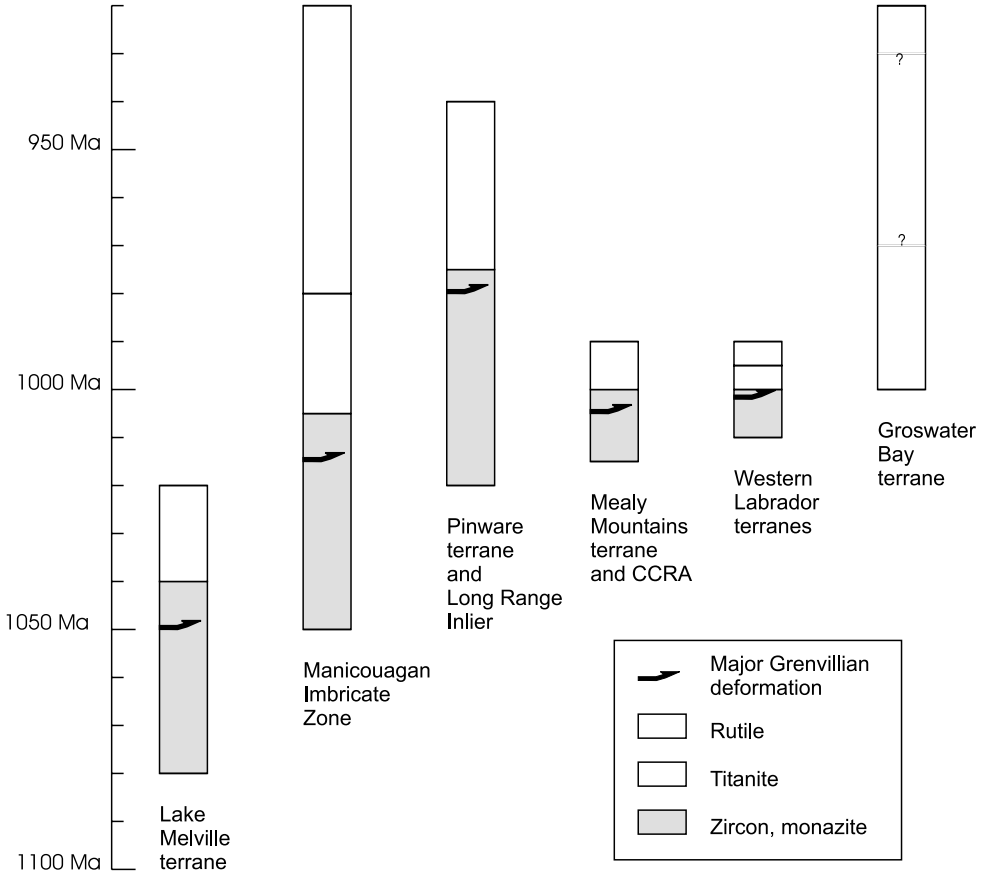
One other minor unit deserves mention at this juncture, namely the east-trending Aillik dykes in the Makkovik Province in eastern Labrador. Fahrig and West (1986) and Buchan and Halls (1990) assigned an age of 1140 Ma to the Aillik dykes, based on (*i*) K–Ar radiometric dating that suggest an age greater than 1000 Ma, and (*ii*) paleomagnetic data that provided a paleopole analytically identical to that from the precisely dated (1141 Ma) Abitibi dykes in Ontario, which have a similar trend and tectonic position with respect to the Grenville front.

Grenvillian orogenesis (1080–985 Ma)

Grenvillian orogenesis in the eastern Grenville Province

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Fig. 13. Time-space diagram illustrating contrasting times of orogenesis, as reflected by zircon, monazite, titanite and rutile ages, and constraints on time of major deformation.



has traditionally been considered to extend from 1080 to 970 Ma (e.g., Gower 1996; but revised slightly here to 985 Ma). This time span is partly predicated on the lack of much evidence for older tectonometamorphic events that can be assigned to the more broadly defined term "Grenvillian orogenic cycle" (Moore and Thompson 1980), which, by one of the more generous time spans offered in the literature, extended from 1300 to 950 Ma (Davidson 1995). The term "Ottawan orogeny" is gaining increasing acceptance for the first half of the period addressed in this section, but is a development that we acknowledge without enthusiasm. To us, the most meaningful definition of the term "Grenvillian orogeny" is for the final series of tectonothermal events that, orogen-wide, collectively led to the creation of the Grenville Province in its present structural configuration. Using this definition we adhere to the 1080 to 985 Ma time span mentioned earlier in this paragraph. In tectonic conceptual terms, this period can be regarded as the terminal continent–continent collisional phase of a continental margin that had been variously somnolent to hyperactive over the preceding 800 million years.

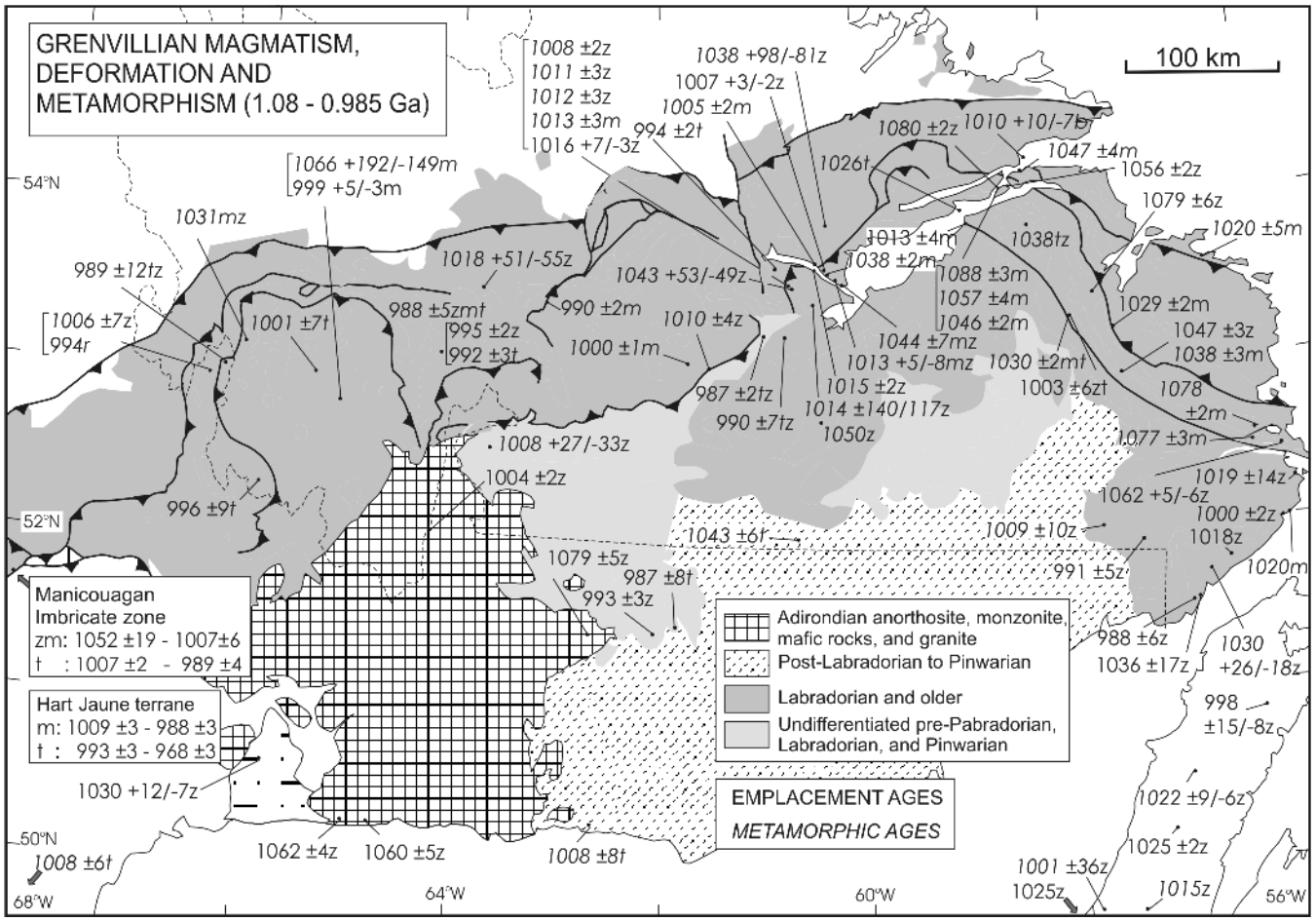
Although some type of Grenvillian deformation and metamorphism was most likely ongoing over the whole Grenville Province between 1080 and 985 Ma, orogenesis peaked in different areas at different times. The following major tectonothermal events are identified, and assigned time ranges, with the end of cooling (to ca. 300°C) given in brackets: (*i*) 1080–1040 Ma (1020 Ma) in the Lake Melville terrane, (*ii*) 1050–1010 Ma (990 Ma) in the Manicouagan Imbricate Zone (immediately west of the region addressed here), (*iii*) 1025–985 Ma (940 Ma) in the Pinware terrane and Long Range Inlier, (*iv*) 1020–1000 Ma (990 Ma) Cape Caribou River allochthon and Mealy Mountains terrane, (*v*) 1010–990 Ma (980 Ma) western Labrador terranes, and (*vi*) 990–970 Ma) in the Groswater Bay terrane (Figs. 13, 14). These are reviewed in the following sections, plus a short section addressing less complete data from southernmost Labrador and adjacent Quebec.

Lake Melville terrane (1080–1040 Ma)

Despite 1080 Ma being taken here as the start of Grenvillian orogenesis in the eastern Grenville Province, one older metamorphic date at 1088 ± 3 Ma from monazite in a metapelite at the west end of Henrietta Island has been reported (Corrigan et al. 2000). If further dating proves that this is not simply an isolated or spurious result, then we acknowledge grounds for expanding the time period. As it is, a zircon ages of 1080 ± 2 Ma for pegmatitic material in a boudinaged neck of a coronitic metagabbro dating deformation (Corrigan et al. 2000) and 1079 ± 6 Ma from the Southwest Brook granite dating emplacement (Schärer and Gower 1988), and monazite determinations from the Rexon's Cove granitic vein at 1078 ± 2 Ma and the Alexis Bay polydeformed granodioritic gneiss at 1077 ± 3 Ma, both dating metamorphism (Scott et al. 1993), convincingly demonstrate tectonothermal activity throughout the Lake Melville terrane at 1080 Ma.

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Fig. 14. Distribution of U–Pb ages during the Grenvillian period. Zircon, z; monazite, m; titanite, t; rutile, r.



The middle phase of the same event is recorded by 1062^{+5}_{-6} Ma (zircon) dating emplacement of the Mecklenburg Harbour granitic vein (Scott et al. 1993); 1057 ± 4 Ma and 1046 ± 2 Ma for the West Henrietta Island metapelite; 1047 ± 4 Ma for the south side of Groswater Bay metapelite (all monazite and dating metamorphism; Corrigan et al. 2000); 1056 ± 2 Ma for the East Henrietta Island late-tectonic pegmatite (zircon dating emplacement; Corrigan et al. 2000); and 1047 ± 3 Ma and 1038 ± 3 Ma from the Paradise River quartz diorite dyke (zircon–emplacement and monazite–metamorphism, respectively; Wasteneys et al. 1997).

Late stages of the event are best known from the west end of Lake Melville, as demonstrated by 1044 ± 7 Ma leucosome from the Northwest River gneiss (monazite and zircon; Krogh 1986), a consistent but very imprecise result of 1038^{-98}_{-81} Ma from the Sebaskatchu tonalite (zircon; Schärer et al. 1986), and 1038 ± 2 Ma from an ultramylonite gneiss at the thrust boundary between the Lake Melville terrane and the overlying Cape Caribou River allochthon (Philippe et al. 1993). Possibly a zircon lower intercept age of 1043^{+53}_{-49} Ma from an amphibolite (Bussy et al. 1995) should be included here, but doing so carries the tacit implication that Lake Melville terrane metamorphism extended to the west side of the Cape Caribou River allochthon. Unfortunately, inadequate precision of the date precludes meaningful development of this thought.

The end–aftermath of the event is inferred from titanite dates (in two cases regressed with zircon or monazite) of ca. 1038 Ma from the Upper North River amphibolite and syenite, 1030 ± 2 Ma from the Second Choice Lake migmatitic gneiss, and ca. 1026 Ma from the Neveisik Island megacrystic granitoid (all from Schärer et al. 1986).

Two determinations from the neighbouring Hawke River terrane are mentioned at this juncture. These are an emplacement age of 1029 ± 2 Ma from the Beaver Brook microgranite dyke (monazite; Schärer et al. 1986) and a metamorphic titanite age of 1020 ± 5 Ma from the 1645 Ma Cartwright alkali-feldspar granite (Kamo et al. 1996). Recall that Labradorian titanite ages from the Hawke River terrane indicate that this region almost entirely escaped Grenvillian high-grade metamorphism. Both of the dating sites mentioned here are near the margins of the Hawke River terrane, which must inevitably have experienced some thermal effects from the adjacent terranes.

Geochronological data demonstrate several important points, enumerated below.

(1) There was no voluminous magmatism in the Lake Melville terrane during early Grenvillian orogenesis; the results date either minor granitoid intrusions, metamorphic events, or are linked to deformation. One exception might seem to be the Southwest Brook granite, which Schärer and Gower (1988) described as a 14-km-wide pluton, but Gower et al.

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(1991) clarified that it was likely a stockwork of granitic veins — indeed, the dated rock came from one such vein. The sporadic magmatism that occurred throughout the period was probably local, linked to the prevailing high-grade conditions.

(2) All three titanite ages postdate 1040 Ma; earlier ages being based on zircon and monazite. That the titanite results fall within a specific period was first recognized by Schärer et al. (1986), who provided the first quantitative data substantiating distinction between the Lake Melville and Groswater Bay terranes. Schärer et al. (1986) regarded the period between 1040 and 1030 Ma as being a short-lived metamorphic, anatectic event, but it is now clear that it reflects closure after a 40-million-year period of tectonism. It should be noted that the older zircon and monazite dates both span the entire pre-1040 Ma period, suggesting that high-grade conditions were maintained throughout the period, rather than any progressive decline in temperature. The late-tectonic pegmatite of Corrigan et al. (2000) and the shallowly discordant granodiorite dyke of Wasteneys et al. (1997) indicate that most deformation was over at those localities by 1056 Ma and 1046 Ma, respectively. It may have been earlier or later elsewhere in the Lake Melville terrane, however (see later in the text).

(3) As discussed by Gower et al. (1996), geochronological data imply temperatures were above ca. 700°C until 1040 Ma and above ca. 600°C until 1030 Ma (closure temperatures of monazite and titanite, respectively; Heaman and Parrish 1991). This introduces difficulties into the conclusion of Scott et al. (1993) that greenschist-facies fabrics are truncated by the Rexon's Cove granitic vein, inferred to have been emplaced syntectonically at 1113 Ma. Given the evidence for widespread, continuous high-grade conditions between 1080 and 1040 Ma, it is difficult to envisage how older low-grade assemblages would be preserved. Gower et al. (1996) suggested that this vein is not syntectonic and that both the apparently truncated fabric and the parallel fabric within the vein were formed later.

Manicouagan Imbricate Zone and Hart Jaune terrane (1050–1010 Ma)

This area is outside the region considered by us to be eastern Grenville Province, but its close proximity, relevance, and the availability of detailed, high-quality geochronological data justify inclusion here. The Manicouagan Imbricate Zone comprises the Lelukuau and Tshenukutish terranes (with the latter subdivided into smaller structural segments), but as a common Grenvillian metamorphic history appears to apply to all parts, geochronological data (Indares et al. 1998; Cox et al. 1998) are reviewed here without spatial reference beyond being part of either the Manicouagan Imbricate Zone or the Hart Jaune terrane.

The date that best establishes the start of Grenvillian tectonism in the Manicouagan Imbricate Zone is 1046 ± 3 Ma metamorphic zircon from a migmatitic amphibolite, taken in preference to, but acknowledging, less precise zircon lower intercept dates of 1052 ± 19 Ma, 1049 ± 22 Ma, 1047 ± 28 Ma, and 1042^{+22}_{-28} Ma from Labradorian gabbro, anorthosite, mangerite and nelsonite, respectively.

A gabbro stock was emplaced slightly later at 1039 ± 2 Ma. Inasmuch as this is the only mafic rock emplaced in the Manicouagan Imbricate Zone during Grenvillian tectonism (all other intrusive rocks being granitic), the age is anomalous. Nevertheless, the data are concordant and precise, so there seems no reason to challenge its veracity, hence special consideration should be accorded to the result, although not a line of inquiry pursued here.

Continuing metamorphism is indicated by metamorphic zircon at 1034 ± 5 Ma from a 1300 Ma leucogranite, 1030 ± 12 Ma from a 1643 Ma hornblendite (baddeleyite included with zircon in the regression), 1030^{+10}_{-7} Ma from a 1170 Ma metagabbro, and 1021^{+54}_{-69} Ma from a 1214 Ma anorthosite. Some associated granitic magmatism occurred, as demonstrated by zircon emplacement ages of 1017 ± 2 Ma from the Hart Jaune granite and 1015 ± 2 Ma from a concordant granitic sheet intruding the Hart Jaune shear zone. Two further zircon results of 1012 ± 12 Ma from a 1170 Ma metagabbro and 1007 Ma from a 1692 Ma? anorthosite demonstrate ongoing metamorphism. This was accompanied by minor granitoid activity, based on a 1007 ± 2 Ma granitic vein (including titanite in the regression) discordantly intruding the Hart Jaune shear zone. The latter age is the oldest in the Manicouagan Imbricate Zone involving titanite and signals the start of declining metamorphism. This is consistent with the observation of Indares et al. (1998) that the 1015 Ma and 1007 Ma dates from concordant and crosscutting granitic sheets in the Hart Jaune shear zone bracket the time of extensional shearing. It is also the justification for the lower time boundary for high-grade metamorphism adopted above.

Early cooling is documented by a titanite ages of 1006 ± 5 and 997 ± 3 Ma from the aforementioned 1046 Ma migmatitic amphibolite (titanite in garnet and in the matrix, respectively), 1006 ± 2 Ma from a 1458 Ma diorite, 1004 ± 4 Ma from a garnet amphibolite, and 991 ± 5 Ma from the 1300 Ma leucogranite. Later cooling is documented by rutile ages of 960 ± 6 Ma, 946 ± 2 Ma (both from the 1046 Ma migmatitic amphibolite), 929 ± 4 Ma from the 1629 Ma gabbro and 920 ± 4 Ma from the 1648 Ma mangerite (the 920 Ma date is from Gale et al. 1994; all others are from Indares et al. 1998 or Cox et al. 1998).

Geochronological data constraining Grenvillian tectonism in the Hart Jaune terrane and the areas to the south is based on monazite dates of 1011 ± 3 Ma, 992 ± 2 Ma, and 989 ± 3 Ma and titanite dates of 992 ± 3 Ma and 967 ± 3 Ma, all from granitic veins. The overall context of these data, only reported in preliminary form by Scott and Hynes (1994*b*), is difficult to evaluate, but no geochronological contrast with the Manicouagan Imbricate Zone is obvious.

From the perspective of this review, the key point demonstrated from the Manicouagan Imbricate Zone geochronological data is its analogy with the Lake Melville terrane. In both areas, metamorphism and deformation involved high pressures and temperatures, were associated with only trivial granitoid magmatism, and had about the same duration. The major difference is in timing, in that tectonism started in the Manicouagan Imbricate Zone when major tectonism ended in the Lake Melville terrane (i.e., at 1050 Ma).

Pinware terrane and Long Range Inlier (1025–985 Ma)

Granitoid plutonism in the Long Range Inlier in Newfound-

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land was synchronous with the later stages of high-grade metamorphism in the Manicouagan Imbricate Zone. Results (preliminary zircon data; Heaman et al. 1996), include emplacement ages of 1025 ± 2 Ma for the Lake Michel granite and, based on one analysis, 1025 Ma for the Lomond River granodiorite. That magmatism was extensive is further suggested by incomplete data that may indicate slightly later emplacement of the Potato Hill charnockite and Apsy granite (ca. 1015 Ma). A concordant zircon age of 1022^{+9}_{-6} Ma from an amphibolite layer within a basement supracrustal package reliably dates the time of granulite-facies metamorphism. This result is augmented by an imprecise result of 1001 ± 36 Ma, which is interpreted as dating metamorphism of 1503 Ma granite from east of Western Brook Pond. A preliminary result of 998^{+15}_{-8} Ma, embodying data from the Horse Chops and Cloud River granites, is considered to date their emplacement and implies a younger phase of granitic magmatism (Heaman et al. 1996).

The Strait of Belle Isle, which conceals some major Phanerozoic faults, divides the Pinware terrane from the Long Range Inlier, but the two regions may be addressed jointly in a Proterozoic context as Cambrian and Ordovician platformal sediments lie unconformably on Grenvillian basement in both areas, and the consensus is that no appreciable structural breaks separate the two. Geochronological data (Tucker and Gower 1994; Wasteneys et al. 1997) indicate that the onset of Grenvillian metamorphism in the Pinware terrane was about the same as that in the Long Range Inlier (i.e., 1025 Ma), based on lower intercept zircon ages of (*i*) 1036 ± 17 Ma from the L'Anse-au-Loup pegmatite that dates the time of its emplacement as well as the metamorphism of its Labradorian volcaniclastic host, (*ii*)1030^{+26}_{-18} Ma from the Labradorian Country Cat monzonite, (*iii*) 1019 ± 14 Ma from The Lodge quartz monzodiorite gneiss, and (*iv*) an imprecise monazite lower intercept of ca. 1020 Ma from the (>1500 Ma) Pleasure Cove quartzite.

Evidence for continued high-grade metamorphism is provided by zircon dates of 1009 ± 10 Ma from the Labradorian "tributary of St. Pauls River" quartz monzonite, 1000 ± 2 Ma from the St. Peter Bay mafic dyke, and monazite dates of 982 ± 5 Ma and 979 ± 20 Ma from the Labradorian Country Cat Pond quartz monzonite and L'Anse-au-Loup volcaniclastic units, respectively. In contrast to the Lake Melville and Manicouagan regions, high-grade metamorphism was accompanied by significant magmatism in the form of the "east of Upper St. Paul River" aegerine syenite emplaced at 991 ± 5 Ma, the Picton Pond quartz monzonite at 988 ± 6 Ma, and the Upper Beaver Brook granite at 983 ± 3 Ma. The last major deformation is bracketed between the very strongly deformed 991 Ma aegerine syenite and the weakly deformed 985 ± 6 Ma L'Anse-au-Loup alkalic mafic dyke (the nominal age for the latter is possibly 5–10 million years too old — see later in the text).

As with the Lake Melville and Manicouagan areas, the time of the last major deformation is linked to the onset of declining metamorphism, as demonstrated by titanite ages (as a result of cooling following uplift). These are 972 ± 5 Ma from the Country Cat Pond gneissic quartz monzonite, 964 ± 12 Ma from the Cape Charles quartz monzonite, 960 ± 6 Ma from The Lodge quartz monzodiorite gneiss, 960 ± 5 Ma from the Upper Beaver Brook granite, 947 ± 10 Ma from the "Tributary of St. Pauls River" quartz monzonite, and 939 ± 5 Ma from the "west of Trout River" quartz monzonite. The span of titanite ages is longer than that in the Lake Melville and Manicouagan areas; we attribute this to the emplacement of late- to post-Grenvillian granitoid plutons that caused higher temperatures to be sustained over a longer period.

Cape Caribou River allochthon and Mealy Mountains terrane (1020–1000 Ma)

The Cape Caribou River allochthon (CCRA) and the Mealy Mountains terrane are treated collectively, following established interpretation that the CCRA is a merely a tectonic outlier, separated from the Mealy Mountains terrane by the Neoproterozoic Lake Melville rift system (e.g., Wardle et al. 1990). Most of the geochronological data comes from the CCRA, and further studies are needed to evaluate the Grenvillian tectonic history of the interior part of the Mealy Mountains terrane. Data from the Lake Melville – Mealy Mountains terrane boundary are also addressed, operating on the hypothesis that the CCRA and Mealy Mountains terrane behaved as a single crustal entity. The mylonitic base to both the CCRA and the Mealy Mountains terrane has been termed the Grand Lake thrust system (Philippe et al. 1993).

Neglecting a result of 1050 Ma of unknown precision from a Pinwarian granite well south of the fringes of the Mealy Mountains terrane (James et al. 2000) and an imprecise date of 1043^{+53}_{-49} Ma (Bussy et al. 1995) from an amphibolite in the Lake Melville terrane (mentioned earlier in the section on Grenvillian metamorphism), geochronological evidence currently suggests a rather short-lived tectonothermal event from 1020 to 1000 Ma. The oldest precise age obtained so far is 1016^{+7}_{-3} Ma (zircon), dating emplacement of a posttectonic pegmatite on the northwest side of the CCRA. A monazite date of 1013 ± 3 Ma from the same pegmatite, and metamorphic zircon ages of 1012 ± 3 Ma, 1011 ± 3 Ma, and 1008 ± 2 Ma from earlier rocks at the same locality (dating amphibolite gneiss, a metadiabase dyke, and leucosome in granitoid gneiss, respectively) provide no indication of earlier Grenvillian tectonothermal activity. This led Bussy et al. (1995), who reported the data, to conclude that the event was very short-lived. Despite this reasoning, it is important to bear in mind that this deformation zone may have been active earlier, coeval with orogenesis in the western Lake Melville terrane (see earlier mention of ca. 1040 Ma dates). This could explain why a 1016 Ma pegmatite appears posttectonic, when most geochronological evidence indicates deformation was later.

The thesis of short-lived orogenesis is supported by data from three closely spaced sites on the east side of the CCRA, where a pre-Labradorian quartz diorite gneiss and a mylonitic paragneiss have yielded metamorphic zircon lower intercept zircon ages of 1015 ± 2 Ma and 1005 ± 2 Ma, respectively, (Philippe et al. 1993), and a pegmatite, regarded as syn-thrusting, has a zircon emplacement age of 1007^{+3}_{-2} Ma (Corrigan et al. 1997). In the same region, a monazite date of 1013^{+5}_{-8} Ma from a Labradorian quartz monzonite records the same metamorphism in the immediate footwall. About 100 km to the southwest, also forming part of the Grand Lake thrust system, a pegmatite, considered to

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be late-thrusting, has a zircon emplacement age of 1010 ± 4 Ma (Corrigan et al. 1997). Within the CCRA, the 1625 Ma Northwest River anorthosite gives an imprecise, but nominally consistent, zircon result of 1014^{+140}_{-117} Ma (Bussy et al. 1995).

On the northeast flank of the Mealy Mountains terrane, long interpreted to be a strike-slip fault, a monazite date of 1013 ± 4 Ma from a pegmatite in the English River shear zone (Corrigan et al. 2000) and a zircon-titanite regression of 1003 ± 6 Ma from the late-tectonic Second Choice Lake pegmatite farther southeast (Gower et al. 1991) provide strong evidence for displacement coeval with that determined for the CCRA. It should be clarified that both of these sites have been previously assigned to the Lake Melville terrane, but, given the Lake Melville terrane's older Grenvillian metamorphic history and the close proximity of these sites to the Mealy Mountains terrane boundary, we have no hesitation in including them here.

The presence of titanite in the 1003 Ma Second Choice Lake pegmatite is consistent with titanite ages of 990 ± 7 Ma and 987 ± 2 Ma from Lower Brook domain leucosome in granitic gneiss and quartz monzonite, respectively, (Philippe et al. 1993). These dates effectively mark the start of declining temperatures around 1000 Ma. We concur with Corrigan et al. (1997) that thrusting was around 1010 Ma, adding our view that this initiated cooling. Thrusting may have been slightly earlier or later in different places, if the syn- or posttectonic status assigned to specific dated pegmatites by various field workers is to be respected.

Western Labrador terranes (1010–990 Ma)

The Lac Joseph, Wilson Lake, Molson Lake, and Churchill Falls terranes are treated collectively here and referred to as western Labrador terranes because current geochronological data provides no indication for any contrasts in Grenvillian tectonothermal history between individual regions. Interestingly, this coherence is also evident in their pre-Grenvillian character, both in lithological make-up and timing of events (demonstrating a long-lived cohesion that justifies the approach undertaken here?). The Lac Joseph and Wilson Lake terranes have been interpreted as thrust-bound allochthonous slices, tectonically emplaced on the Molson Lake and Churchill Falls terranes. The latter two terranes are really one structural entity with separate names for its western and eastern parts, as has been stressed by James (1994).

A key point that should be stated at the outset is that the Grenvillian metamorphism in the Lac Joseph and Wilson Lake terranes was comparatively mild (analogous to parts of the Mealy Mountains terrane in eastern Labrador). Many of the U–Pb geochronological determinations made on samples from the Lac Joseph terrane do not show any Grenvillian Pb loss in zircon, and, in those that do, analytical points plot so close to upper intercepts as to render lower intercepts meaninglessly imprecise. Two examples are a 1066^{+192}_{-149} Ma result from an N2 leucosome (1636 Ma) and 1031 Ma from a mylonitized migmatite (1633 Ma) having unspecified, but large error limits (Connelly and Heaman 1993). In addition, several of the pre-Grenvillian ages in these terranes (see earlier) are defined by monazite and titanite, which denies a granulite-facies Grenvillian overprint. One other imprecise result is a 1018^{+51}_{-55} Ma zircon date (Brooks 1983) from the Sona Lake granitoid gneiss (1649 Ma) in the Churchill Falls terrane. This is derived from a pooled regression of two related samples and should be regarded with caution.

From the perspective of commencement of Grenvillian metamorphism in the western Labrador terranes, none of the above dates provide rigorous constraints. The oldest that does is a zircon lower intercept age of 1006 ± 7 Ma from a Shabogamo gabbro in the Molson Lake terrane, which also contains concordant rutile at 994 Ma (Connelly and Heaman 1993). Synchronous ages of 1001 ± 7 Ma from a 1660 Ma granite (titanite; Connelly et al. 1995) and 1000 ± 1 Ma from a pelitic diatexite terrane (monazite, Corrigan et al. 1997) have been obtained from the Lac Joseph and Wilson Lake terranes, respectively. Late-tectonic pegmatites having emplacement monazite ages of 999^{+5}_{-3} Ma from the Lac Joseph terrane (Connelly and Heaman 1993) and 990 ± 2 Ma from the Wilson Lake terrane (Corrigan et al. 1997) provide a younger limit to the time of major deformation. Other determinations are all from Labradorian granitoid rocks and are 996 ± 9 Ma and 989 ± 12 Ma from the Molson Lake terrane (Connelly and Heaman 1993) and 995 ± 2 Ma, 992 ± 3 Ma, and 988 ± 5 Ma from the Churchill Falls terrane (Krogh 1994). All involve titanite, although zircon is included in some of the regressions. The only other result is a less precise baddeleyite date of 966 ± 30 Ma from the Shabogamo gabbro exposed in a tectonic window through the base of the central Lac Joseph terrane (Connelly et al. 1995).

The conclusion that has been drawn from these essentially synchronous ages on minerals with very different blocking temperatures is that the emplacement of the allochthonous terranes over their footwall crust was very rapid (Krogh 1994; Connelly et al. 1995). It would seem that metamorphism did not get underway much before 1010 Ma, and, based on the dates from late-tectonic pegmatites, thrusting ended by 1000 Ma. Taking the 994 Ma rutile as a guide, cooling was also extraordinarily rapid. Such alacrity certainly distinguishes Grenvillian metamorphism in the western Labrador terranes from other parts of the eastern Grenville Province.

Groswater Bay terrane (990 Ma? – 970 Ma)

Grenvillian dates from the Groswater Bay terrane show a very different pattern to that evident in all the other regions described in the preceding sections. In contrast to the model of high-grade metamorphism, then thrusting followed by cooling, current data only indicate the final phase. This may be due to inadequate studies, but might genuinely reflect the tectonic signature of this region, noting that Pb-loss in zircon is relatively minor in most of the (Labradorian) rocks that make up the terrane.

The only geochronological straw to cling to regarding significant Grenvillian metamorphism is a lower-intercept result of 1010^{+10}_{-7} Ma based on baddeleyite and zircon from a sample that was assigned to Michael gabbro by Corrigan et al. (2000). Concerns that the upper intercept of 1472 Ma is "too old" were outlined earlier. A similar situation may well apply to the lower intercept. If data from the Michael gabbro obtained by Schärer et al. (1986) is included in the regression, the line would be rotated to younger upper and lower intercepts, which might be more meaningful.

With two exceptions, all the remaining Grenvillian ages are based on titanite, although zircon is included on the regression line in two cases. The ages are 994 ± 2 Ma from

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migmatitic gneiss at the west end of the Groswater Bay terrane (Philippe et al. 1993), 980 ± 4 Ma from the Bluff Head hornblende granodiorite, 978 ± 4 Ma from the Double Island granodiorite to tonalite gneiss, 977 ± 2 Ma from the Saddle Island metasedimentary gneiss (all from Schärer et al. 1986), 975 ± 27 Ma from the Susan River quartz diorite (Philippe et al. 1993), 973 ± 11 Ma from the Grady Island intrusion (Kamo et al. 1996), 972 ± 4 Ma from the Cuff Island nebulitic granodiorite, ca. 970 Ma from the Double Island pegmatite, and 968^{+7}_{-8} Ma from the Cuff Island banded granodiorite gneiss (all from Schärer et al. 1986). The ca. 970 Ma date from the Double Island pegmatite is a slightly discordant Pb–Pb titanite age based on one analytical point, but time of closure cannot have been very different from the 978 Ma date obtained from its host. Of the two exceptions, one is an imprecise date of 941^{+39}_{-36} Ma from the Double Mer granite based on a long extrapolation from zircon analyses plotting close to the upper intercept (Schärer et al. 1986). The other is rutile from a gabbro at Cuff Island, which yielded nearly concordant results of 931 Ma and 923 Ma, hence establishing the end of Grenvillian effects in the Groswater Bay terrane.

The thesis that Grenvillian metamorphism was comparatively mild in the Groswater Bay terrane seems in conflict with a north-to-south systematic increase in metamorphism to high-grade metamorphic assemblages in the Michael gabbro that are comparable with those in its Labradorian hosts. As the 1426 Ma Michael gabbro postdates Labradorian metamorphism and there is no evidence for post-emplacement, pre-Grenvillian high-grade metamorphism, it would seem that the severe overprint must be Grenvillian. An escape from this dilemma was proposed by Gower and Erdmer (1988), whereby the high-grade assemblages in the Michael gabbro are attributed to subsolidus (isobaric?) cooling after intrusion. They postulated that the nature of the various coronitic assemblages depended on the depth at which reequilibration occurred; gabbro that reached mid- to high-crustal levels only shows relatively moderate-grade assemblages, whereas that emplaced closer to the base of the crust developed a high-grade mineralogy. Following isostatic adjustments, the (now exposed) crust remained cool enough to escape Grenvillian high-grade conditions, but was reconfigured by stepped thrusts that exhumed progressively deeper crustal levels from north to south and brought about titanite closure. Note that this thesis is consistent with the model of Gower et al. (1997*b*) that the Lake Melville – Mealy Mountains terrane boundary represents the limit of Grenvillian allochthonous terranes and that the region to the northeast escaped this overthrusting.

Southernmost Labrador and adjacent Quebec

Geochronological data demonstrating Grenvillian tectonothermal activity in the southern part of the eastern Grenville Province, excluding the eastern Pinware terrane, are too sparse to attempt any meaningful synthesis, but are included here for completeness.

The oldest Grenvillian ages in the southern part of the eastern Grenville Province identify an intrusive and metamorphic event in the Havre St. Pierre anorthosite massif region, based on results from four samples. From the Romaine River augen monzogranite, Loveridge (1986) determined an age of 1079^{+17}_{-11} Ma (but expressed uncertainty as to whether the zircons were igneous or metamorphic). Close to the coast, van Breemen and Higgins (1993) reported an age of 1062 ± 4 Ma from pegmatitic anorthosite, but, in fact, two separate samples 20 km apart were analysed, having concordant Pb–Pb ages of 1062 ± 4 Ma and 1060 ± 5 Ma. It is worth observing that, although the body sampled has been previously grouped with the 1128–1126 Ma Havre St. Pierre anorthosite, magnetic maps show it to be a separate intrusion mostly situated offshore in the Gulf of St. Lawrence.

Farther northeast, the oldest reported age is 1043 ± 6 Ma (titanite; James et al. 2001) from a 1493 Ma K-feldspar megacrystic granitoid rock assigned to the Mecatina terrane. The 1043 Ma age is interpreted to date metamorphism, but its context relative to other Grenvillian events in the same area remains uncertain. Somewhat younger, and closer to the Havre St. Pierre anorthosite, the Bigot granite yielded a zircon age of 1030^{+12}_{-7} Ma, which is interpreted to date emplacement (Gobeil et al. 1999). The closest contemporary intrusions to this date just mentioned are the 1025 Ma granitoid intrusions in the Long Range Inlier. The emplacement was also coeval with pre-thrusting high-grade metamorphism in the Manicouagan Imbricate Zone and post-thrusting cooling in the Lake Melville terrane.

Metamorphism is imprecisely dated to be 1008^{+27}_{-33} Ma by lower-intercept zircon from the 1225 Ma Lac Long tonalite to quartz diorite gneiss (Krogh 1985) and 1008 ± 8 Ma titanite from the 1246 Ma Davy Group porphyry (Clark and Machado 1995). Immediately west of the eastern Grenville Province, anorthositic mylonite from the Rivière Pentecôte intrusion has yielded an age of 1008 ± 6 Ma (Martignole et al. 1993). Emplacement of the Lac Brigeart granite at 1004 ± 2 Ma (zircon; Krogh and Schärer 1987) and the Lac Galissonière quartz monzonite at 993 ± 3 Ma (zircon; Loveridge 1986) may be manifestations of a significant ca. 1000 Ma Grenvillian magmatic event coeval with thrusting farther north in the western Labrador terranes. Declining temperatures are shown by titanite metamorphic ages of 987 ± 8 Ma and 981 ± 4 Ma from the Natashquan River and Lac Brigeart granites, respectively, (Martignole et al. 1994; Krogh and Schärer 1987).

Late Grenvillian magmatic events

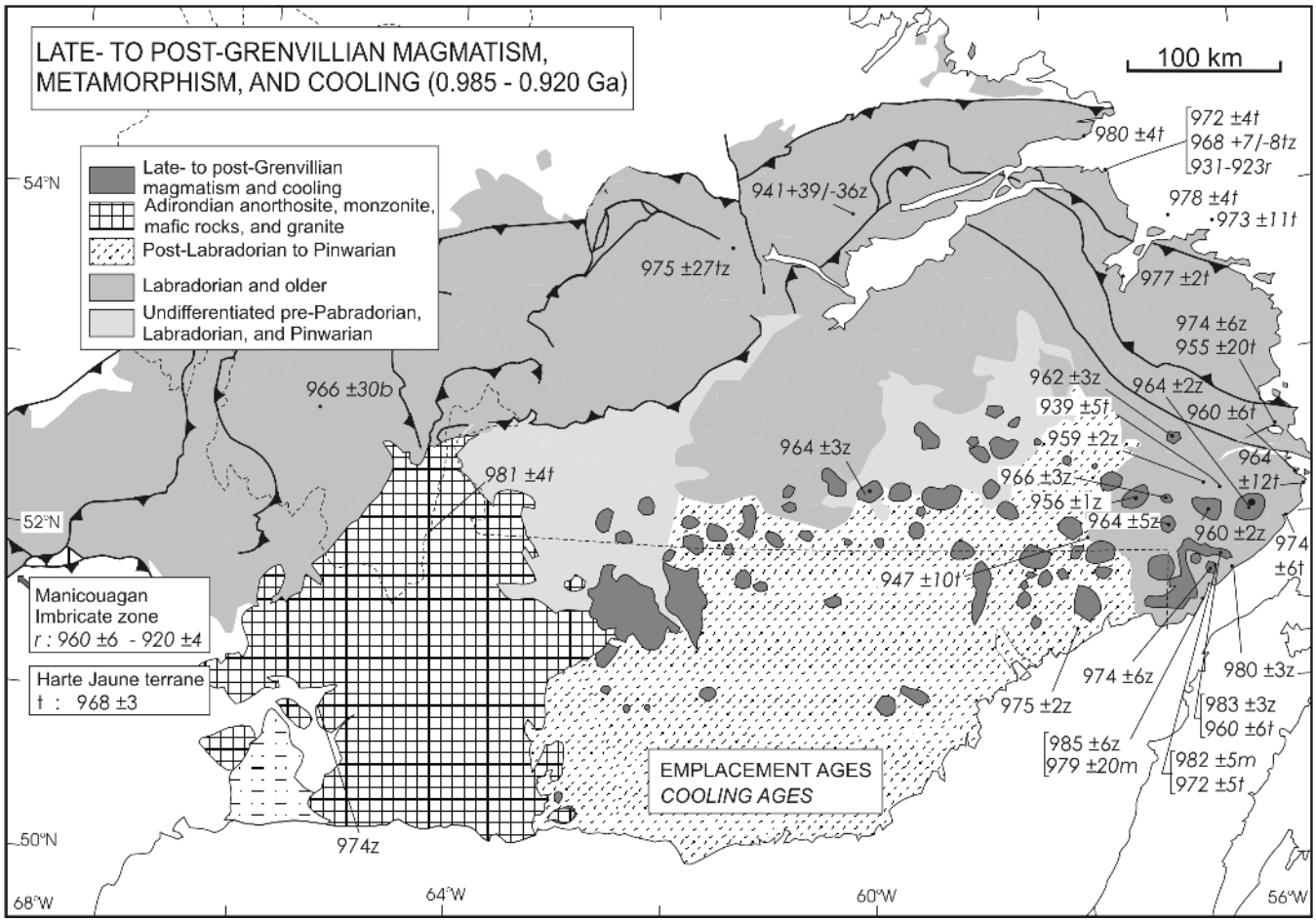
Late Grenvillian events are divided here into two groups (cf. Fig. 15). The older event (985 to 975 Ma) was characterized by alkalic mafic dyking and anorthositic–alkalic magmatism, whereas the younger event (975–955 Ma) was one of monzonitic, syenitic, and granitic magmatism. The first clearly developed into the second without any hiatus, but we feel there are sufficient time–compositional contrasts to merit reviewing them separately below.

Early posttectonic magmatism (985–975 Ma)

Alkalic mafic dykes are widespread in the Pinware terrane and have also been recognized in the Lake Melville terrane, where they are spatially associated with a major brittle facture zone (Gilbert River fault). The dykes discordantly intrude their host rocks, are overall east-southeast-trending (although commonly irregular and branching), in some cases are extremely xenolithic, and may show magma-mingling characteristics (Gower et al. 1994). The southernmost dykes

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Fig. 15. Distribution of igneous rocks and U–Pb ages during late- to post-Grenvillian events. Zircon, z; monazite, m; titanite, t; rutile, r.



show the most mineralogical modification and were emplaced at deeper levels than the pristine northern dykes, which were intruded close to the surface (as indicated by amygdales in the mafic intrusions and clastic dykes in the Gilbert River fault). Two emplacement zircon ages have been obtained of 985 ± 6 Ma from the L'Anse-au-Diable dyke in the south (Pinware terrane) and 974 ± 6 Ma from the Gilbert Bay dyke in the north (Lake Melville terrane) (Wasteneys et al. 1997). A titanite cooling age of 955 ± 20 Ma was also reported from the Gilbert Bay dyke by Wasteneys et al. (1997). Although precision limits permit coeval emplacement, the nominally older age obtained from the southern dyke is consistent with its mildly metamorphosed state, having been injected during waning Grenvillian metamorphism in the area. Thus it may be considered Grenvillian structurally, but post-Grenvillian magmatically. Note that the L'Anse-au-Loup alkalic mafic dyke discordantly intruded the 1036 Ma pegmatite mentioned earlier, and is, itself, crosscut by an undated, sheared aplite.

Major intrusive bodies emplaced at the same time include the Red Bay gabbro (Gower et al. 1994; Greenough and Owen 1995), dated to be 980 ± 3 Ma (Heaman et al., in preparation[2]), the 975 ± 2 Ma Vieux-Fort granophyric leuconorite (Perreault and Heaman 2001), and the 974 ± 6 Ma fayalite-bearing Lower Pinware River alkali-feldspar syenite (Heaman et al., in preparation). The latter date supercedes an earlier result of 962^{+76}_{-120} Ma obtained by Wasteneys et al. (1997), who argued on the basis of other geochronological data and field evidence that emplacement was between 985 and 972 Ma.

Late posttectonic magmatism (975–955 Ma)

Late posttectonic magmatism can be classified as either minor granitic intrusions, up to a few metres across, or plutons several kilometres in diameter.

Only two dates have been reported for minor granitoid intrusions, but such rocks are undoubtedly widespread. The ages are 974 ± 6 Ma from the St. Peter Bay aplite (Tucker and Gower 1994), based on titanite, and 974 Ma (error unspecified) for a microgranite dyke from east of Lake Manitou (Gobeil et al. 1999). Neither date can be considered rigorous; Tucker and Gower (1994) cautioned that their result, being based on titanite, could reflect an event after emplacement (e.g., metamorphism), and Gobeil et al.'s (1999) date relies on one analytical point (albeit nearly concordant).

The granitoid plutons are discrete, circular to elliptical (in plan) bodies; range in composition between monzonite, syenite, and granite; are typically homogeneous and massive; and

[2] Heaman, L.M., Gower, C.F., and Perreault, S. Refinements in the understanding of Labradorian, Pinwarian and Grenvillian evolution of the Pinware terrane in southeasternmost Labrador and adjacent Quebec (in preparation).

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locally show mantled feldspar textures. They are commonly associated with distinct positive aeromagnetic anomalies. Recognition of these intrusions as representing a major addition of granitic magma to the eastern Grenville Province was first made by Gower et al. (1991), who, from limited knowledge of their distribution at the time, used aeromagnetic patterns and meagre reconnaissance geological mapping to predict that they would be found to be extensive and concentrated in the southern half of the eastern Grenville Province. The names "Exterior Thrust Belt" and "Interior Magmatic Belt" were proposed to distinguish conceptually the northern and southern regions (Fig. 1). Subsequent mapping has confirmed that many (but not all) of the magnetic anomalies correlate with late posttectonic granitoid plutons. Few of the more recently mapped bodies have been studied geochronologically, so the number of known intrusions far exceeds those dated. This is primarily because such intrusions can be confidently identified in the field, which has meant that geochronological resources have been redirected elsewhere. Ages determined are 966 ± 3 Ma for the Upper St. Lewis River (east) monzonite, 964 ± 3 Ma for the Joir River granite, 964 ± 2 Ma for the Chateau Pond granite, 964 ± 5 Ma for the Riviere Bujeault headwaters quartz syenite, 962 ± 3 Ma for the Southwest Pond granite, 960 ± 2 Ma for the Upper Pinware River granite, 959 ± 2 Ma for the Upper Trout River granite, and 956 ± 1 Ma for the Upper St. Lewis River (west) granite. All ages are based on concordant or near-concordant arrays of zircon analyses that were reported by Gower and Loveridge (1987), Gower et al. (1991), Tucker and Gower (1994), or James et al. (2001).

Post-Grenvillian waning (955–920 Ma)

On the basis of results from rutile from various parts of the eastern Grenville Province (see earlier in the text), tectonothermal activity waned between 955 and 920 Ma. After this, the region subsided into a well-earned repose after 900 million years of mountain building, moving and destroying. This rest lasted 300 million years until the dawn of the opening of Iapetus Ocean, which was marked in the Grenville Province by the emplacement of the Long Range dykes dated to be 615 ± 2 Ma (Kamo et al. 1989) and 615^{+6}_{-4} Ma (Kamo and Gower 1994) and accompanied by deposition of the Double Mer Formation in the Lake Melville Rift System and Sandwich Bay graben.

Tectonic model

Elzevirian orogenesis (1230–1180 Ma)

It was emphasized earlier that no geochronological data exist that we consider reliably dates any event in the eastern Grenville Province between 1230 and 1180 Ma. Some evidence from the Wakeham Supergroup hints at Elzevirian deformation, and, outside our self-imposed U–Pb mandate, Ar–Ar data also suggest a thermal disturbance at this time. More to the point, however, is that tectonic conditions clearly changed around this time, inasmuch as factors ceased that had allowed the emplacement of AMCG suites, alkalic felsic magmatism, mafic dyking and associated extrusive activity and sedimentation that had prevailed for more than 200 million years during the Elsonian. The last dated Elsonian events were felsic magmatism at 1240 Ma in the north and south (Strange Lake alkaline complex and Natashquan River granites, respectively) and mafic magmatism at 1224 Ma in the middle (Naskaupi sills).

We interpret the change in tectonic setting to be a consequence of accretion of outboard island-arc terranes farther to the southwest, now preserved as the Central Metasedimentary Belt and comparable rocks in mid- and southern-U.S.A. Whether similar accretion occurred south of Pinwarian crust in the eastern Grenville Province remains unknown, but a consequence of the accretion was to move the zone of active subduction outboard of the newly accreted crust, while imposing a compressive tectonic regime in eastern Laurentia that brought Elsonian conditions to a close.

Post-Elzevirian – pre-Grenvillian (1180–1080 Ma)

Gower (1996) suggested that AMCG and mafic magmatism between 1180 and 1120 Ma represent the distal expression of a north-dipping subduction zone in a back-arc mixed extension and compression environment, drawing parallels with tectonic conditions during geon 14. Although concurring broadly with that comparison, we feel that some refinements are required. Unlike the 1460 to 1290 Ma period, there is little evidence of a time migration of activity relative to the inferred continental margin. Magmatism occurred along a broad front and was accompanied by deformation, rather than being funnelled flat subduction as hypothesized for Elsonian magmatism.

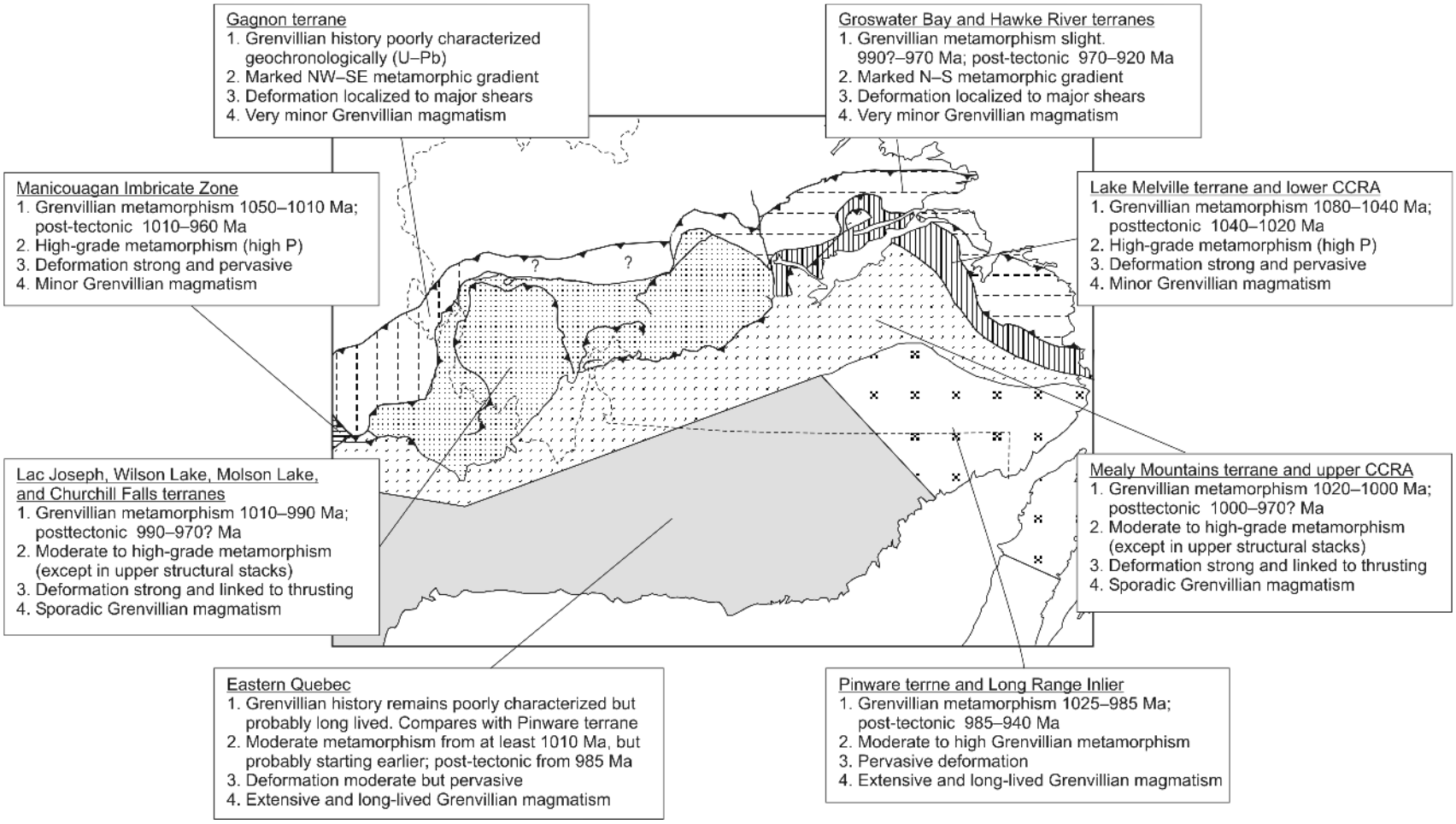
Grenvillian (1080–980 Ma)

Although Grenvillian deformation and metamorphism was most likely ongoing over the whole Grenville Province between 1080 and 980 Ma, we offer some new postulates for the eastern Grenville Province. These are that (*i*) orogenesis peaked in different areas at different times, (*ii*) as the crust yielded through thrusting in one place, stresses were transferred elsewhere, and (*iii*) coeval emplacement of large igneous bodies occurred in southern areas separate from those subjected to high-grade conditions. Note that these concepts are quite different from the proposal that there were time-limited, orogen-wide pulses at various stages of Grenvillian orogenesis (e.g., Rivers 1997; Rivers and Corrigan 2000). The first two postulates are taken to mean that there was periodic shifting of the "pressure point" responsible for compressional tectonism during Grenvillian collisional orogenesis. This was not so evident in the south where a different set of conditions prevailed, in which fluids played a more important role. In the following discussion, we retain the concept that the Grenville Province can be subdivided into an Exterior Thrust Belt and an Interior Magmatic Belt (Gower et al. 1991). Armed with much more data than a decade ago, the observation holds valid that the Exterior Thrust Belt experienced minor and sporadic magmatism, in contrast with the Interior Magmatic Belt, where it was voluminous. Key tectonic elements are summarized in Fig. 16.

All regions in the Exterior Thrust Belt passed through the same three orogenic stages. The first stage, as demonstrated by zircon and monazite ages, lasted up to 40 million years and culminated in thrusting (Fig. 13). What little magmatism there was occurred during this stage and was related to high-grade conditions, rather than emplacement from a distant source. This was followed by a 10- to 20-million-year-long middle stage that was marked by cooling due to exhumation after thrusting. This period is demonstrated by titanite ages,

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Fig. 16. Summary of key features of blocks of terranes. Note similarity between paired terrane groups from top to bottom. CCRA, Cape Caribou River allochthon.



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which may be interpreted to reflect initial cooling from high-grade conditions. The final stage was one of continued cooling, probably without much activity, and is demonstrated by rutile ages. Data are not adequate anywhere to establish its duration, but it may have lasted up to 60 million years. These three stages all had about the same duration in the Lake Melville terrane, the Manicouagan Imbricate Zone, and the Pinware – Long Range region, with stages 1 and 2 being about 30– 40 million years and 20 million years, respectively. For the remaining regions, these stages were much shorter, about 20 million years and 10 million years, respectively.

Orogenesis commenced first in the Lake Melville terrane and, about 30 million years later, in the Manicouagan Imbricate Belt. Note that activity commenced in the Manicouagan Imbricate Belt at about the same time that most deformation in the Lake Melville terrane ceased (ca. 1050 Ma). Similarly, orogenesis appears to have started in the Pinware – Long Range region at about the time it was over in the Manicouagan area (ca. 1020 Ma). There is less evidence for large age differences in timing of orogenesis between the Mealy Mountains – CCRA block, Lac Joseph – Wilson Lake – Molson Lake – Churchill Falls block and Groswater Bay terrane, but neither can it be assumed that events were exactly synchronous in all three regions. What does seem likely, however, is that events proceeded much faster. We consider that it was these later deformational events that are responsible for out-of-sequence thrusting, at the same time overriding and concealing regions that were deformed during the earlier stages of Grenvillian orogenesis, with the exception of the Lake Melville terrane and Manicouagan Imbricate Zone. The final phase of tectonism was uplift along the Grenville front, which we assume occurred along its full length despite supporting U–Pb evidence being currently confined to the Groswater Bay terrane.

One modification to the Interior Magmatic Belt concept as originally proposed by Gower et al. (1991) is that magmatism probably occurred throughout Grenvillian orogenesis, rather than being concentrated at the posttectonic stage. Evidence is not compelling as yet that this was the case, so the claim relies on (*i*) the fact that all Grenvillian plutons dated since 1991 are in the southern half of the eastern Grenville Province, (*ii*) a wider representation of ages has been obtained from them than previously established, and, (*iii)* in contrast to the north, there are numerous undated candidates. In all cases, the plutons were emplaced into preexisting crust, with no evidence for juvenile material.

Posttectonic Grenvillian (985–955 Ma)

The weakly deformed and slightly recrystallized state of ca. 980 Ma intrusions in the Pinware terrane and the pristine state of post-975 Ma across the southern part of the eastern Grenville Province provides one measure of the end of Grenvillian orogenesis. Enough evidence is available to distinguish between an early posttectonic AMCG–mafic–alkalic magmatic event and late posttectonic monzonite–syenite–granite plutons. The simplest model is to envisage them to be the consequence of crustal thickening following collision, but we leave it to another publication to address this issue more insightfully.

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